

Neural attentional filters and behavioural outcome follow independent individual trajectories over the adult life span

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Abstract

Preserved communication abilities promote healthy aging. To this end, the age-typical loss of sensory acuity might in part be compensated for by an individual's preserved attentional neural filtering. Is such a compensatory brain-behaviour link longitudinally stable? Can it predict individual change in listening behaviour? We here show that individual listening behaviour and neural filtering ability follow largely independent developmental trajectories modelling electroencephalographic and behavioural data of $N = 105$ aging individuals (39–82 yrs). First, despite the expected decline in hearing-threshold-derived sensory acuity, listening-task performance proved stable over 2 years. Second, neural filtering and behaviour were correlated only within each separate measurement timepoint (T1, T2). Longitudinally, however, our results raise caution on attention-guided neural filtering metrics as predictors of individual trajectories in listening behaviour: Neither neural filtering at T1 nor its two-year change could predict individual two-year behavioural change, under a combination of modelling strategies.

eLife assessment

This study provides a **valuable** contribution to understanding the neural mechanisms underlying age-related changes in attention and speech understanding. The large dataset ($N=105$) provides **convincing** evidence for how speech recognition behaviour and neural tracking of speech separately evolve in about 2 years. The work would be of interest to psychologists, neuroscientists, and audiologists.

Introduction

Speech comprehension is an essential aspect of human communication, enabling us to understand and interact with others effectively. Preserved communication is therefore critical to social well-being and healthy aging^{1,2}. Any translational advance aimed at maintaining and restoring

successful cognitive aging crucially relies on understanding the factors that explain and predict individual trajectories of listening performance. However, the evidence on these potential factors is astonishingly limited.

As we age, our ability to comprehend speech can decline due to age-related changes in the auditory system (i.e., sensory acuity) and in cognitive resources. Age-related hearing loss reduces the ability to detect and to discriminate speech sounds, especially in noisy environments. However, it has long been recognised that increasing age and hearing loss cannot fully account for the considerable degree of inter-individual differences observed in listening behaviour and its lifespan change^{2,4}.

Recent research has focused on the neurobiological mechanisms that promote successful speech comprehension by implementing ‘neural filters’ that segregate behaviourally relevant from irrelevant sounds. Such neural filter mechanisms act by selectively increasing the sensory gain for behaviourally relevant inputs or by inhibiting the processing of irrelevant inputs^{5–7}. A growing body of evidence suggests that speech comprehension is neurally supported by an attention-guided filter mechanism that modulates sensory gain and arises from primary auditory and perisylvian brain regions: By synchronizing its neural activity with the temporal structure of the speech signal of interest, the brain ‘tracks’ and thereby better encodes behaviourally relevant auditory inputs to enable attentive listening^{8–11}.

In a large age-varying cohort (N = 155; 39–80 yrs), we have previously shown how the fidelity of this neural filtering strategy can help explain differences in listening behaviour (i) from individual to individual, and (ii) within individuals from sentence to sentence^{12–14}. As participants performed a challenging dual-talker listening task, we recorded their electroencephalogram (EEG). We observed that enhanced neural filtering—defined as stronger neural tracking of attended vs. ignored speech—led to more accurate and overall faster responses. Notably, we observed both neural filtering as well as its link to behaviour to be independent of chronological age and severity of hearing loss¹² (**Fig. 1A**).

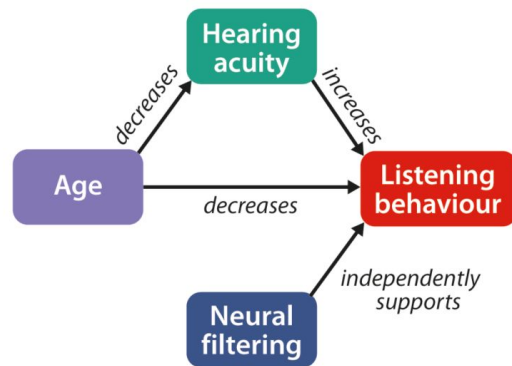
The observation of such brain–behaviour relationships critically advances our understanding of the neurobiological foundation of cognitive functioning by showing, for example, how neural implementations of auditory selective attention support attentive listening. They also provide fruitful ground for scientific inquiries into the translational potential of neural markers. However, the potency of neural markers to predict future behavioural outcomes is often only implicitly assumed but seldomly put to the test¹⁵.

Using auditory cognition as a model system, we here aim to overcome this limitation by testing directly the hitherto unknown longitudinal stability of neural filtering as a neural compensatory mechanism upholding communication success. Going further, we ask to what extent an individual’s attentional neural-filtering ability measured at a given moment is able to predict their future trajectory in listening performance. Only if this is the case, and only if such an association can plausibly be assumed to be causal for future changes in communication ability, neural filtering would be a potential translational target.

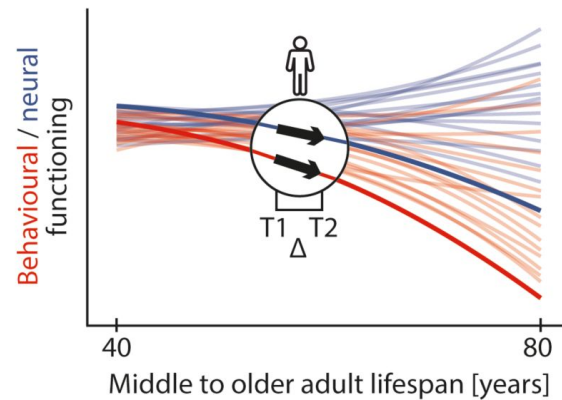
We here aim to fill this gap by analysing two-year changes in the sensory, neural, and behavioural domain in a longitudinal subsample (N = 105; 39–82 yrs) of the original AUDADAPT cohort¹² (see **Fig. 2A**). We apply a combination of advanced cross-sectional and longitudinal modelling strategies to address the following specific research questions (see **Fig. 1**).

First, by focusing on each domain individually, we ask how sensory, neural and behavioural functioning evolve cross-sectionally across the middle and older adult life span (**Fig. 1B**). More importantly, we also ask how they change longitudinally across the studied two-year period (**Fig. 1C**, Q1), and whether aging individuals differ significantly in their degree of change (Q2). We

A Determinants of listening behaviour



B Analysis of individual change



C Explaining and predicting neural and behavioural change

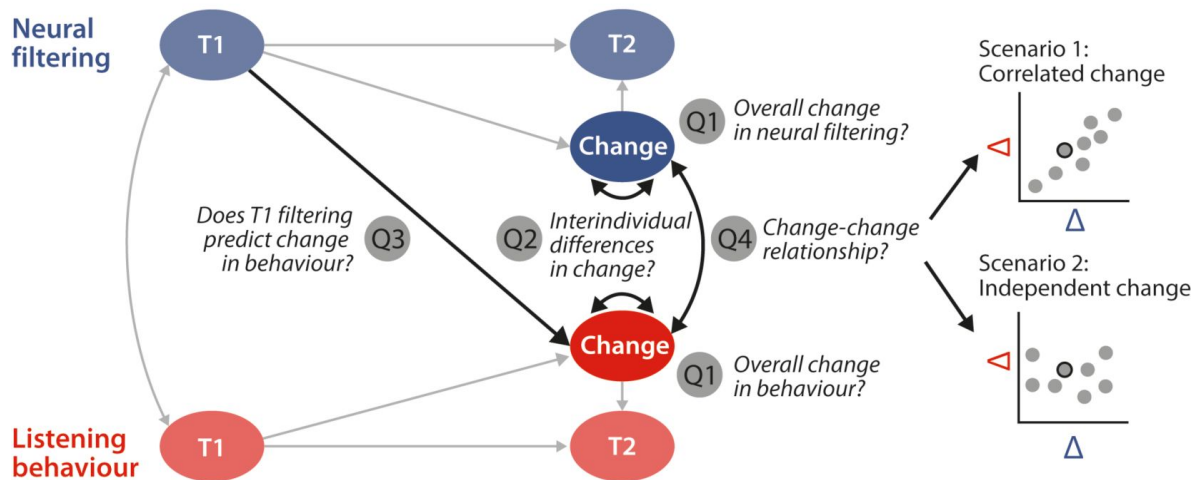


Figure 1.

Schematic illustration of key assumptions and research questions. A

Listening behaviour at a given timepoint is shaped by an individuals' sensory and neural functioning. Increased age decreases listening behaviour both directly, and indirectly via age-related hearing loss. Listening behaviour is supported by better neural filtering ability, independently of age and hearing acuity. **B** Conceptual depiction of individual two-year changes along the neural (blue) and behavioural (red) domain. Thin coloured lines show individual trajectories across the adult lifespan, thick lines and black arrows highlight two-year changes in a single individual. **C** Left, Schematic diagram highlighting the key research questions detailed in the introduction and how they are addressed in the current study using latent change score modelling. Right, Across individuals, co-occurring changes in the neural and behavioural domain may be correlated (top) or independent of one another (bottom).

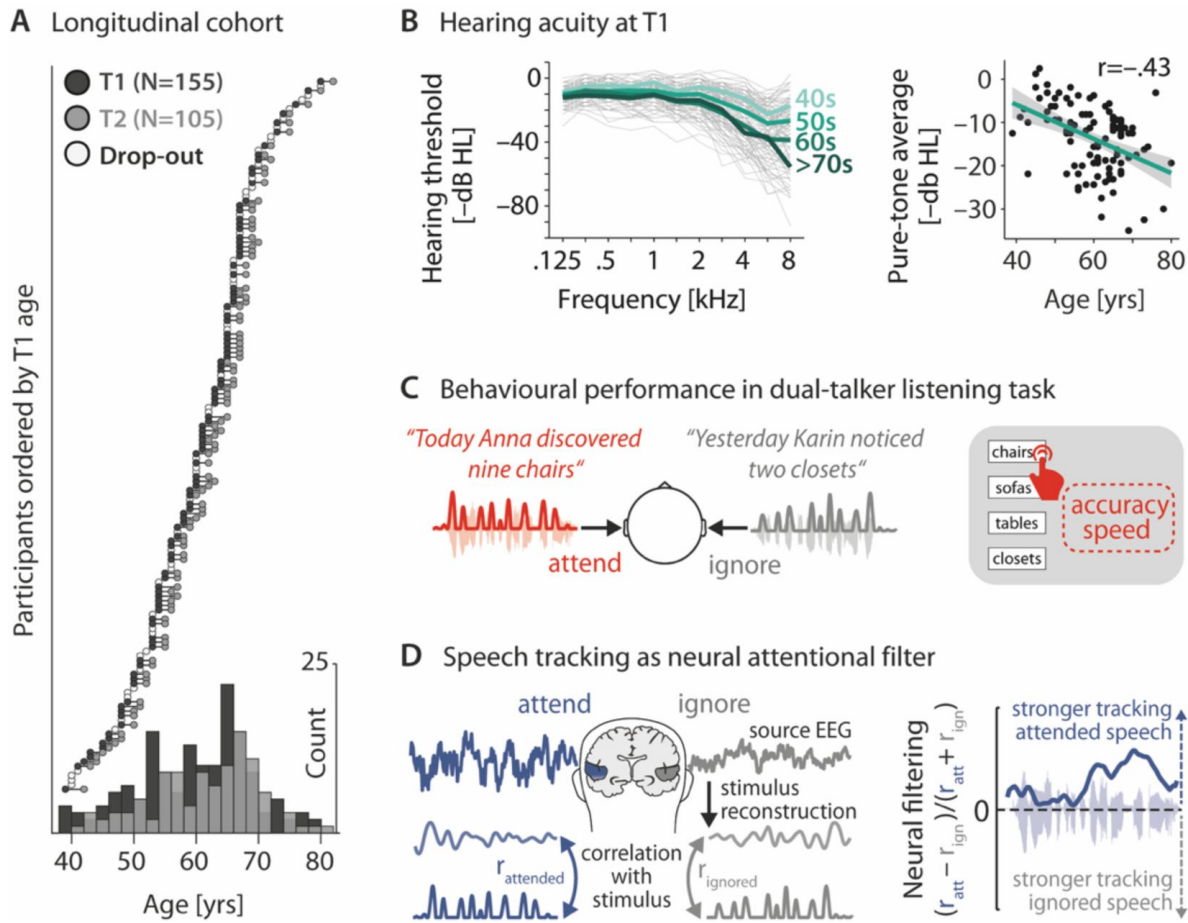


Figure 2.

Key neural and behavioural metrics derived from a longitudinal cohort. A

Longitudinal cohort of healthy middle-aged and older adults measured twice, two years apart. Circles represent individual participants at a given measurement time (dark grey: timepoint (T) 1, light grey: T2, white: drop-outs after T1). Bottom, Age distribution at T1 and T2 across 5-year bins. **B** Left, T1 air conduction hearing thresholds per individual (thin grey lines) and age group (thick coloured lines). Note that for didactic purposes, throughout the manuscript, thresholds are expressed as -dB HL to highlight the decrease in hearing acuity with age (left). Right, Pure-tone average hearing acuity (0.5, 1, 2, and 4 kHz across both ears; higher is better) negatively correlates with age ($r = -.43$, $p = 3.73 \times 10^{-6}$). **C** Participants listened to two sentences presented simultaneously to the left and right ear. In 50 % of trials, a preceding visual cue indicated the to-be-attended target sentence. Listening behaviour is quantified via the accuracy and speed in identifying the final word of the target sentence. **D** Left, neural speech tracking as a proxy of an individual's neural filtering ability. Stimulus envelopes of attended and ignored sentences were reconstructed from source-localized EEG activity in auditory cortex (see Methods for details) and correlated with the actual envelopes. Right, Better neural filtering results from stronger neural tracking of attended compared to ignored speech. We analysed neural filtering derived from the entire sentence presentation period.

expect individuals' hearing acuity and behaviour to decrease from T1 to T2. Since we previously observed inter-individual differences in neural filtering to be independent of age and hearing status, we did not expect any systematic longitudinal change in neural filtering.

Second, we test the longitudinal stability of the previously observed age- and hearing-loss-independent effect of neural filtering on both accuracy and response speed (**Fig. 1A**). To this end, we analyse the multivariate direct and indirect relationships of hearing acuity, neural filtering and listening behaviour within and across timepoints.

Third, leveraging the strengths of latent change score modelling^{16,17}, we fuse cross-sectional and longitudinal perspectives to probe the role of neural filtering as a precursor of behavioural change in two different ways: we ask whether an individual's T1 neural filtering strength can predict the observed behavioural longitudinal change (Q3), and whether two-year change in neural filtering can explain concurrent change in listening behaviour (Q4). Here, irrespective of the observed magnitude and direction of T1–T2 developments, two scenarios are conceivable: Intra-individual neural and behavioural change may be either be correlated—lending support to a compensatory role of neural filtering—or instead follow independent trajectories¹⁸ (see **Fig. 1C**).

Answering these questions is vital for understanding the neurobiological mechanisms of successful communication across the lifespan. Answering them will also critically inform the development of interventions targeted at maintaining or restoring communication success and therefore concerns basic and applied researchers alike.

Results

We studied an age-varying cohort of healthy middle-aged and older adults longitudinally (N = 105, 39–82 yrs, median age at T2: 63 yrs). The mean time difference between the two measurement timepoints reported here was 23.2 (sd: 4.0) months. We characterise the multivariate relationship of key measures of sensory, neural, and behavioural functioning to explain and predict individual trajectories of listening performance.

At each of two measurement timepoints, participants underwent audiological assessment followed by EEG recording during which they performed a difficult dual- talker dichotic listening task (see **Fig. 2A,B**).^{12,19,20} In each trial of the task, participants listened to two temporally aligned but spatially separated five-word sentences. They then had to identify the final word in one of the two sentences from a visual array of four alternatives given a 4-sec time limit (**Fig. 2C**). In 50 % of the trials, a visual spatial- attention cue indicated the side of target sentence presentation, the other half of trials were preceded by an uninformative neutral cue.

We extracted individuals' mean accuracy and response speed (calculated as the inverse of reaction time) as key readouts of listening behaviour. On the basis of source- localised 1–8 Hz auditory cortical activity, we further quantified individuals' neural filtering ability as their attention-guided neural tracking of relevant vs. irrelevant speech (**Fig 2D**; see also ref.¹² and Methods for details). Our main analyses focus on neural filtering and listening performance averaged across all trials and thereby also across two separate spatial-attention conditions. This choice allowed us to most directly probe the trait-like nature and relationships of neural filtering. It was additionally supported by our previous observation of a general boost in behavioural performance with stronger neural filtering, irrespective of spatial attention¹².

We follow a three-step analysis strategy to address our specific research questions: First, we provide a largely descriptive overview of the observed average two- year change per studied domain. Second, we follow up on this fundamental analysis with a causal mediation analysis²¹.

and single-trial mixed-effects model analysis geared to assess the longitudinal stability of our recently reported effects of age, hearing acuity and neural filtering on listening task performance. Third and final, we integrate and extend the first two analysis perspectives in a joint latent change score model (LCSM)¹⁶ to most directly probe the role of neural filtering ability as a predictor of future attentive listening ability. Addressing our key change-related research questions at the latent rather than the manifest level supersedes the manual calculation of notoriously noisy differences scores, and effectively removes the influence of each metric's reliability on the estimation of change-related relationships^{16,17,22}.

Listening performance remains stable despite decreased hearing acuity

In a first analysis (**Fig. 3**), we characterised how hearing acuity, neural filtering, and listening performance change across the middle to older adult lifespan. Additionally, we analysed longitudinal change from timepoint 1 (T1) to timepoint 2 (T2). We used the same linear mixed-effect models to test cross-sectional effects of age and longitudinal changes with time. We additionally quantified each measure's test-retest reliability as their T1–T2 Pearson's correlation.

Note that throughout the manuscript and all analyses, we reversed the sign of pure-tone average (PTA) values to express them as an index of hearing acuity rather than hearing loss (i.e., higher values indicating better acuity). Similarly, for more intuitive interpretation, accuracy is visually presented as mean proportion correct but was logit transformed for all statistical analyses to satisfy model assumptions.

As expected, hearing acuity decreased linearly with increasing age (**Fig. 3A**, $\beta = -3.4$, SE (standard error) = .71, $p < .001$) and on average by 1.2 dB from T1 to T2 ($\beta = -1.18$, SE = .27, $p < .001$; meanT1: -13.72 dB HL (sd: 7.8); meanT2: -14.90 dB HL (8.3)). The effect of age did not change with time (age x timepoint $\beta = -.35$, SE = .28, $p = .21$). Assuming constant individual progression rates, this observed change corresponds to a projected average decrease in hearing acuity per decade of -6.3 (sd:15.3) dB HL (see Fig. S1). The magnitude of observed and projected hearing loss progression is well in line with recent large-sample reports^{23–25}. Measurements of hearing acuity showed high test-retest reliability ($r = .94$, $p < .001$), underscoring the high fidelity of our audiological assessment.

In line with known deleterious effects of age, both behavioural outcomes (response speed and accuracy) declined with increasing age, and did so to a similar degree in T1 and T2 (**Fig. 3C, D**; Speed: $\beta = -.02$, SE = .01, $p = .004$; Accuracy: $\beta = -.23$, SE = .07, $p = .0001$; timepoint x age $p > .12$). At the same time, and contrary to our expectations, average performance levels remained stable from T1 to T2 (Speed: $\beta = .004$, SE = .01, $p = .44$; meanT1: .62 s⁻¹ (.08); meanT2: .62 s⁻¹ (.08); Accuracy: $\beta = .04$, SE = .05, $p = .36$, meanT1: .88 % (.09); meanT2: .88 % (.11)). Accuracy and response speed showed moderately high test-retest reliability (Speed: $r = .70$, $p < .001$; Accuracy: $r = .72$, $p < .001$).

The analysis of change in neural filtering revealed that its strength varied independently of age at both timepoints (**Fig. 3B**, $\beta = .0003$, SE = .0005, $p = .48$; timepoint x age $\beta = -.0002$, SE = .0006, $p = .79$), confirming our previously reported T1 results¹². As shown in **Figure 3** (bottom left panel), magnitude and direction of observed longitudinal change are highly variable across individuals and age groups, and we did not find evidence of any systematic group-level change from T1 to T2 ($\beta = .001$, SE = .001, $p = .16$). In addition, individual neural filtering strength correlated only weakly across time ($r = .21$, $p = .03$).

We also assessed the reliability of two established neural traits using resting-state EEG from the same recording sessions: The individual alpha frequency (IAF)²⁶ and the slope of 1/f neural noise^{27,28}. As expected, both metrics showed high test-retest reliability (IAF: $r = .83$, $p < .001$; 1/f

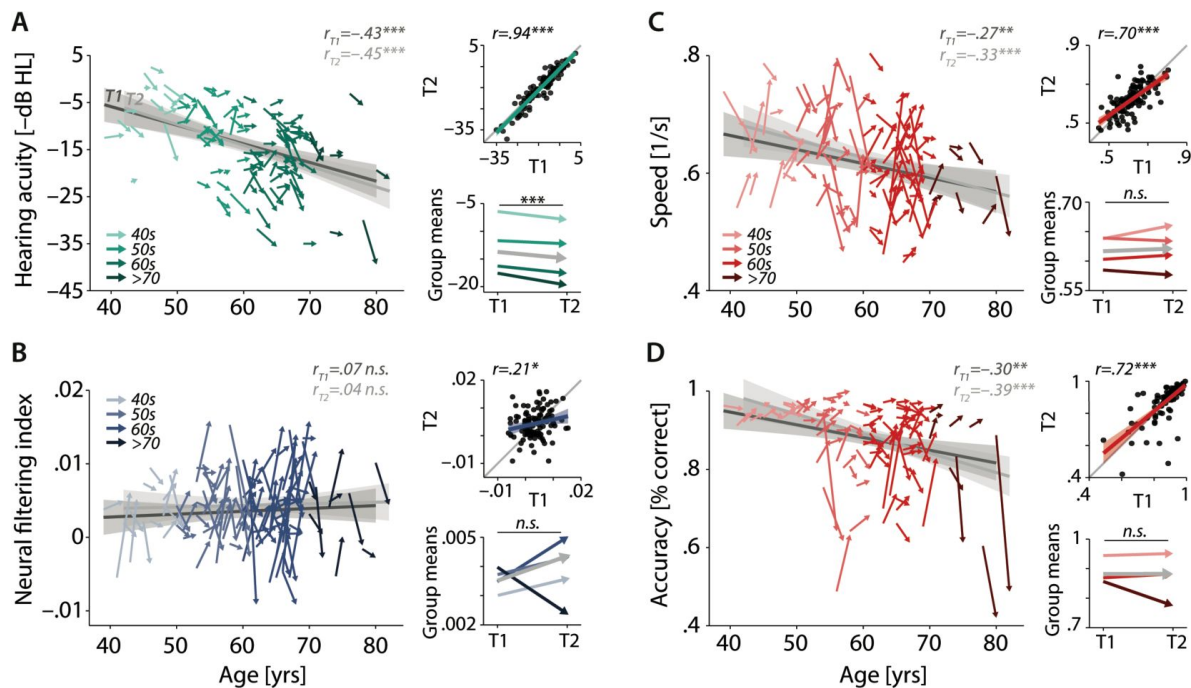


Figure 3.

Characterizing cross-sectional and longitudinal change along the auditory sensory (A), neural (B), and behavioural (C, D) domain. For each domain, coloured vectors (colour-coding four age groups for illustrative purposes, only) in the respective left subpanels show an individual's change from T1 to T2 along with the cross-sectional trend plus 95% confidence interval (CI) separately for T1 (dark grey) and T2 (light grey). Top right subpanels: correlation of T1 and T2 as measure of test-retest reliability along with the 45° line (grey) and individual data points (black circles). Bottom right panels: Mean longitudinal change per age group (coloured vectors) and grand mean change (grey). Note that accuracy is expressed here as proportion correct for illustrative purposes, but was analysed logit-transformed or by applying generalized linear models.

slope: $r = .78$, $p < .001$). These findings provide a reference level on reliability, demonstrating that the weak reliability of the neural filtering metric is not due primarily to differences in EEG signal quality across sessions.

The temporal instability of neural filtering challenges its status as a potential trait-like neural marker of attentive listening ability. At the same time, irrespective of the degree of reliability of neural filtering itself, across individuals it may still be reliably linked to the behavioural outcome (see [Fig. 1](#)). This is being addressed next.

Neural filtering reliably supports listening performance independent of age and hearing status

On the basis of the full (T1 and T2) dataset, we aimed to replicate our key T1 results and test whether the previously observed between-subjects brain-behaviour relationship would hold across time: We expected an individual's neural filtering ability to impact their listening performance (accuracy and response speed) independently of age or hearing status¹². Given the moderately strong correlation of age and hearing acuity ($r = -.43$; $p < .001$; [Fig. 2B](#)), we employed causal mediation analysis to model the direct as well as the hearing-acuity-mediated effect of age on the behavioural outcome²¹. To formally test the stability of direct and indirect relationships across time, we used a moderated mediation analysis. In this analysis, the inclusion of interactions by timepoint tested whether the influence of age, sensory acuity, and neural filtering on behaviour varied significantly across time.

Our expectations on the direct relationships were indeed borne out by the data: higher age was associated with poorer hearing ability ($\beta = -.43$, $SE = .09$, $p < .001$) and listening performance (Speed: $\beta = -.33$, $SE = .06$, $p < .001$; Accuracy: $\beta = -.26$, $SE = .06$, $p < .001$). Better hearing ability, on the other hand, boosted accuracy but not response speed (Accuracy: $\beta = .30$, $SE = .1$, $p = .003$; Speed: $\beta = -.06$, $SE = .1$, $p = 0.56$). These direct effects remained stable from T1 to T2 (all interactions by timepoint $p > .56$; all log Bayes Factors ($\log BF_{01}$) > 2.5).

Age also impacted accuracy indirectly: The total effect of age was partially mediated via its detrimental effect on hearing acuity (average causal mediation effect (ACME), $\beta = -.12$, $SE = .04$, $p < .001$). We did not find evidence for an analogous indirect effect on speed (ACME: $\beta = .008$, $SE = .03$, $p = .77$). Again, the hearing-acuity mediated effect of age on accuracy did not change from T1 to T2 as evidenced by moderated mediation analysis (interaction by timepoint $p = .73$).

Speaking to the robustness of our previous results, we observed the beneficial effect of stronger neural filtering fidelity on both measures of listening performance (Accuracy: $\beta = .21$, $SE = .09$, $p = .02$; Speed: $\beta = .33$, $SE = .09$, $p < .001$). Note that the magnitude of this direct brain-behaviour effect is comparable to that of the direct effect of age. Alternative models that included indirect, neural filtering-mediated paths from either age or hearing acuity to behaviour did not reveal any significant mediation effects.

Most importantly, the longitudinal stability of the observed direct brain-behaviour link was further supported by the absence of any significant changes with time (interactions by timepoint; all $p > .28$, all $\log BF_{01} > 2.1$).

In our previous T1 analysis ($N = 155$)¹², we had found evidence for the here analysed brain-behaviour link at two different levels of observation: (i) at the trait level—individuals with overall stronger neural filtering also performed better overall—and (ii) at the state-level—stronger neural filtering in a given trial raised the chances of responding correctly. Aiming at replication of the state-level (i.e., within-participant) relationship, we ran a single-trial linear mixed-effects model analysis on our longitudinal $N = 105$ sample. This analysis utilised single-trial data of both T1 and T2.

Lending credibility to our previous results, stronger single-trial neural filtering was associated with higher listening success at both T1 and T2 (logistic mixed-effects model; within-participant effect of neural filtering on accuracy: odds ratio (OR) = 1.08, SE = .02, $p < .001$; interaction neural filtering x timepoint: OR = .99, SE = .03, $p = .82$; see also **Fig. 4C**).

Accuracy is longitudinally stable but speed and neural filtering increase at T2

Having established the longitudinal stability of the beneficial impact of intact neural filtering on listening performance, we turned to our final, most comprehensive analysis.

In a latent change score model (LCSM), in its bivariate form sometimes termed a parallel process model^{16,17}, we connected the neural and behavioural domain. This allowed us to most directly probe the potential role of neural filtering as a precursor of behavioural changes. Specifically, we asked: (i) Is an individual's baseline (T1) level of neural filtering ability predictive of their two-year change in behaviour; and (ii) are individual differences in longitudinal dynamics in the behavioural domain associated with those in the neural domain?

As a technical note, it is worth reiterating that in the present data, the highly variable, weakly reliable surface measure of neural filtering was nonetheless robustly connected to the behavioural outcome (see above; see **Fig. 4C**). It is in such scenarios that the LCSM framework comes with particular methodological benefits: By expressing individuals' T1 and T2 levels, as well as their T1–T2 change as latent variables instead of manifest indicators, these types of models circumvent the calculation of notoriously unreliable noisy difference scores. They also avoid potential regression to the mean due to random errors^{17,22}. Instead, the measurement errors of both, latent variables and their associated indicators, are explicitly modelled and thus effectively removed from the estimates of individual differences and relationships of interest¹⁶.

Accordingly, for our metrics, we estimated T1 and T2 latent variables of behavioural and neural filtering from two manifest indicators each. These indicators were the average of each metric across the first and second half of the experiment, respectively (see Methods for details). For all measurement models, the standardized factor loadings were significant (all $p < .05$; all standardized $\alpha > .55$). The assumption of strict factorial invariance across time could be maintained for all models (all $\Delta\chi^2(df = 1) < 3.4$, all $p > .07$).

We then constructed univariate latent change score models (LCSM) to test for significant mean change in each metric from T1 to T2 while adjusting for their respective baseline (T1) level. The univariate models had acceptable (Speed: $\xi^2(df = 5) = 9.7$, $p = .085$, CFI = .988, RMSEA = .094] to excellent fit [Accuracy: $\xi^2(df = 5) = 5.1$, $p = .40$, CFI = .999, RMSEA = .016; Neural filtering: $\xi^2(df = 5) = .6$, $p = .99$, CFI = 1, RMSEA = 0) according to established indices²⁹.

On average, listening task accuracy remained stable ($b_0 = .053$, SE = .045, $\Delta\chi^2(df = 1) = 1.33$, $p = .25$). Response speed, on the other hand, showed a significant mean increase over time ($b_0 = .13$, SE = .03, $\Delta\chi^2(df = 1) = 12.79$, $p < .001$; **Fig. 5A**). Similarly, the model of neural filtering showed a (marginally) significant mean increase ($b_0 = .24$, SE = .11, $\Delta\chi^2(df = 1) = 2.98$, $p = .08$; **Fig. 5A**).

As a proof of principle, we extracted latent factor scores for T1/T2 neural filtering and response speed from their respective univariate latent change score model. By correlating T1 and T2 factor scores per domain, we show how the explicit modelling of measurement errors helps to improve the test-retest reliability compared to that observed at level of manifest variables (see insets in **Fig. 5A**; neural filtering: $r = .65$, $p < .001$, response speed: $r = .75$, $p < .001$).

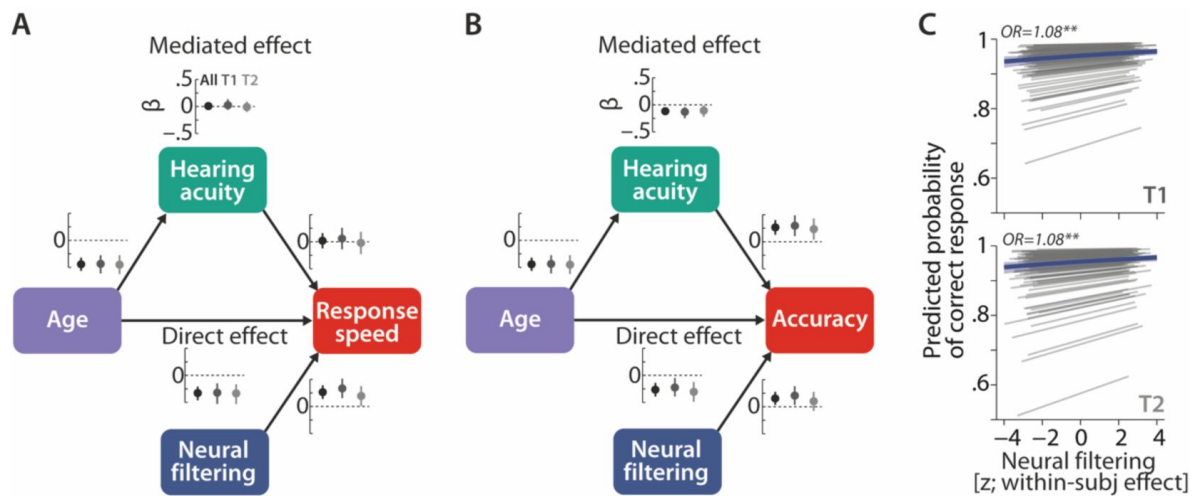


Figure 4.

Longitudinal stability of sensory and neural determinants of listening behaviour.

Causal mediation analysis of age, hearing acuity, and neural filtering on response speed (**A**) and logit-transformed accuracy (**B**). Graphs next to each path indicate standardised coefficients plus 95% CIs separately for the full dataset (black), T1 (dark grey), and T2 (light grey). We did not find any significant modulation of the observed effects with time (see text for results). **C** Neural filtering strength was found to be predictive of accuracy at the single-trial level at both T1 (top) and T2 (bottom). Grey lines show individual effects, blue thick line shows the group-level fixed effect along with 95 % confidence interval. OR = odds ratio, ** $p < .01$.

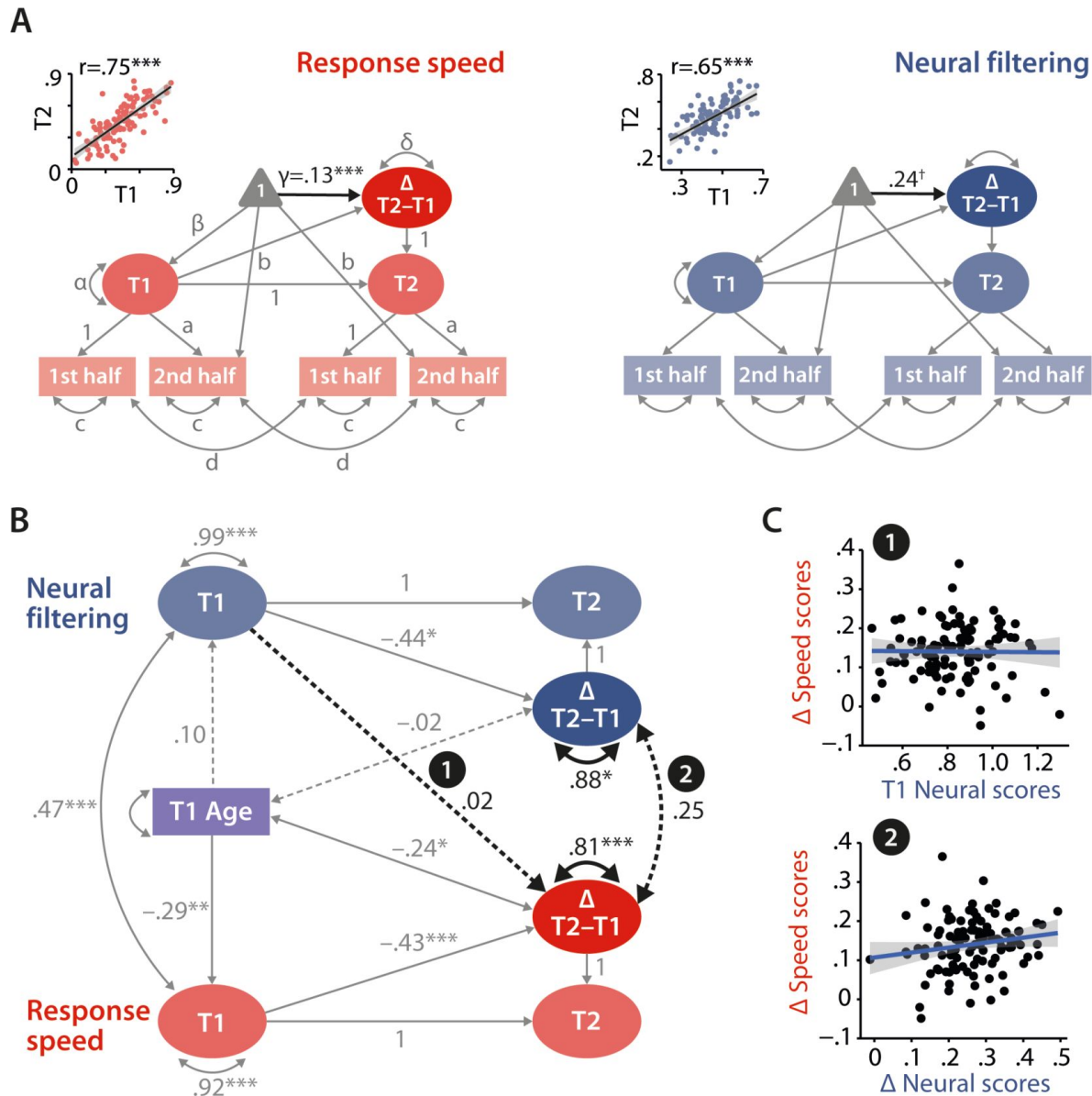


Figure 5.

Modelling of univariate and bivariate change. A

Univariate latent change score models for response speed (left) and neural filtering (right). All paths denoted with Latin letters refer to freely estimated but constrained to be equal parameters of the respective measurement models. Greek letters refer to freely estimated parameters of the structural model. Highlighted in black is the estimated mean longitudinal change from T1 to T2. Scatterplots in the top left corner illustrate how capturing T1 and T2 neural and behavioural functioning as latent factors improves their respective test-retest reliability. **B** Latent change score model (LCSM) relating two-year changes in neural filtering strength to changes in response speed. Black arrows indicate paths or covariances of interest. Solid black arrows reflect freely estimated and statistically significant effects, dashed black arrows reflect non-significant effects. All estimates are standardised. Grey arrows show paths that were freely estimated or fixed as part of the structural model but that did not relate to the main research questions. For visual clarity, manifest indicators of the measurement model and all symbols relating to the estimated mean structure are omitted but are identical to those shown in panel A. *** $p < .001$, ** $p < .01$, * $p < .05$, † $p = .08$. **C** Scatterplots of model-predicted factor scores that refer to the highlighted paths in panel B. Top panel shows that baseline-level neural filtering did not predict two-year change in behavioural functioning, bottom panel shows the absence of a significant change- change correlation.

Baseline neural functioning does not predict future change in listening behaviour

Based on the univariate results, we then connected the two metrics that showed a significant mean change, namely speed and neural filtering, in a bivariate model of change (**Fig. 5B**; see also Fig. S2 for full model details).

In line with our hypotheses, we modelled the longitudinal impact of T1 neural functioning on the change in speed, and tested for a change-change correlation. Since the analyses conducted up to this point have either directly shown or have suggested that longitudinal change per domain may be affected by age, we included individuals' age as a time-invariant covariate in the final model. We modelled the influence of age on neural and behavioural functioning at T1 but also on individual change per domain. By accounting for linear effects of age on longitudinal change, we also minimize its potential impact on the estimation of change-change relationship of interest. Note that we refrained from fitting separate models per age group due to both limited and different number of data points per age group. The model fit the data well ($\chi^2(df = 27) = 25.65$, $p = .54$, CFI = 1, RMSEA = 0, 95 % CI[0 .07]).

Having ensured factorial invariance and goodness of fit, we can confidently interpret the estimates of individual differences and bivariate relationships that speak to our specific research questions. Crucial to our change-related inquiries, we observed reliable variance (i.e., individual differences) in the longitudinal change in both speed (standardized estimate $\phi = 0.88$, SE = .07, $\Delta\chi^2(df = 5) = 94.07$, $p < .01$) and neural filtering ($\phi = .81$, SE = .14, $\Delta\chi^2(df = 4) = 25.64$, $p < .001$).

Individuals' baseline levels of both speed and neural filtering strength were predictive of their respective longitudinal change: Individuals with relatively strong neural filtering or fast responses at T1 showed a smaller increase from T1 to T2, possibly indicating ceiling effects (Speed: $\beta = -.43$, SE = .12, $\Delta\chi^2(df = 1) = 8.75$, $p = .003$; Neural filtering: $\beta = -.44$, SE = .16, $\Delta\chi^2(df = 1) = 4.17$, $p = .04$).

We also observed that participants' age at T1 covaried with the individual degree of change in speed but not with that in neural filtering: The older a participant at T1, the smaller their longitudinal increase in speed (Speed: $\phi = -.24$, SE = .11, $\Delta\chi^2(df = 1) = 4.38$, $p = .037$; Neural filtering: $\phi = -.02$, SE = .14, $\Delta\chi^2(df = 1) = .02$, $p = .89$).

Importantly, however, an individual's latent T1 level of neural filtering strength was not predictive of the ensuing latent T1–T2 change in response speed ($\beta = .02$, SE = .16, $\Delta\chi^2(df = 1) = .02$, $p = .90$). We did not have any *a priori* hypotheses on the influence of T1 speed on the individual T1–T2 change in neural filtering. Still, in a control analysis that freely estimated the respective path, we found that an individual's latent T1 level of response speed was not predictive of the ensuing latent T1–T2 change in neural filtering ($\beta = -.11$, SE = .21, $\Delta\chi^2(df = 1) = .31$, $p = .58$).

Neural filtering ability and listening behaviour follow independent developmental trajectories in later adulthood

Finally, we turn to the last piece in our investigation where we address the question of whether individual differences in the neural and behavioural longitudinal change are connected. In other words: Are the contemporaneous changes along the two studied domains correlated or do they occur largely independently of one another? Change score modelling revealed that longitudinal change in the neural and the behavioural domain occurred largely independent of one another despite their systematic relationship within each separate measurement timepoint ($\phi = .25$, SE = .15, $\Delta\chi^2(df = 1) = 2.74$, $p = .1$). In other words, those individuals who showed the largest change in neural filtering were not necessarily the ones who also changed the most in terms of their behavioural functioning (see **Fig. 5C** bottom panel and Fig. S3).

Control analyses: The weak correlation of behavioural and neural change is robust against different quantifications of neural filtering

Taken together, our main analyses revealed that inter-individual differences in behavioural change could only be predicted by baseline age and baseline behavioural but not neural functioning, and did not correlate with contemporaneous neural changes.

However, one could ask in how far core methodological decisions taken in the current study, namely our focus on (i) the *differential* neural tracking of relevant vs. irrelevant speech as proxy of neural filtering, and (ii) on its trait-level characterization that averaged across different spatial-attention conditions may have impacted these results. Specifically, if the neural filtering index (compared to the neural tracking of attended speech alone) is found to be less stable generally, would this also impact the chances of observing a systematic change-change relationship? Relatedly, did the analysis of neural filtering across all trials underestimate the effects of interest?

To evaluate the impact of these consideration on our main findings, we conducted two additional control analyses: First, we repeated the main analyses using the neural filtering index (and response speed) averaged across selective-attention trials, only. Second, we repeated the main analyses focused on the neural tracking of attended speech only, again averaged across selective-attention trials.

As shown in **Figure 6**, taken together, the control analyses provide compelling empirical support for the robustness of our main results: Linking response speed and neural filtering under selective attention strengthened their relationship at T1 ($\phi = .54$, $SE = .15$, $\chi^2(df = 1) = 2.74$, $p = .1$; see **Fig 6B**) but did not yield any significant effects for the influence of T1 neural filtering on behavioural change ($\beta = .13$, $SE = .21$, $\chi^2(df = 1) = .43$, $p = .51$), or for the relationship of neural and behavioural change ($\phi = .26$, $SE = .14$, $\chi^2(df = 1) = 3.1$, $p = .08$; please note the close correspondence to path estimates reported in **Fig. 5**).

The second control analysis revealed a substantially higher manifest-level test-retest reliability of neural tracking of attended speech ($r = .65$, $p < .001$; **Fig. 6C**) compared to that of the neural tracking index. However, when linked to longitudinal changes in response speed, this analysis provided even less evidence for systematic change-related relationships: Baseline-levels of attended-speech tracking did not predict future change in response speed ($\beta = .18$, $SE = .11$, $\chi^2(df = 1) = 2.73$, $p = .10$), and changes in neural and behavioural functioning occurred independently of one another ($\phi = -.03$, $SE = .12$, $\chi^2(df = 1) = .06$, $p = .81$).

In sum, the two control analyses provide additional empirical support for the results revealed by our main analysis.

Discussion

Successfully comprehending speech in noisy environments is a challenging task, particularly for aging listeners whose hearing ability gradually declines³⁰. A much-researched neural support mechanism here is attention-guided neural ‘tracking’ of behaviourally relevant speech signals, as one neural strategy to maintain listening success^{7–10,14,31–33}.

However, to date, it is unknown if the fidelity with which an individual implements this filtering strategy does represent a stable neural trait-like marker of individual attentive listening ability. Of direct relevance to any future translational efforts building on neural speech tracking, it is also unknown whether differences in neural filtering strength observed between aging listeners are

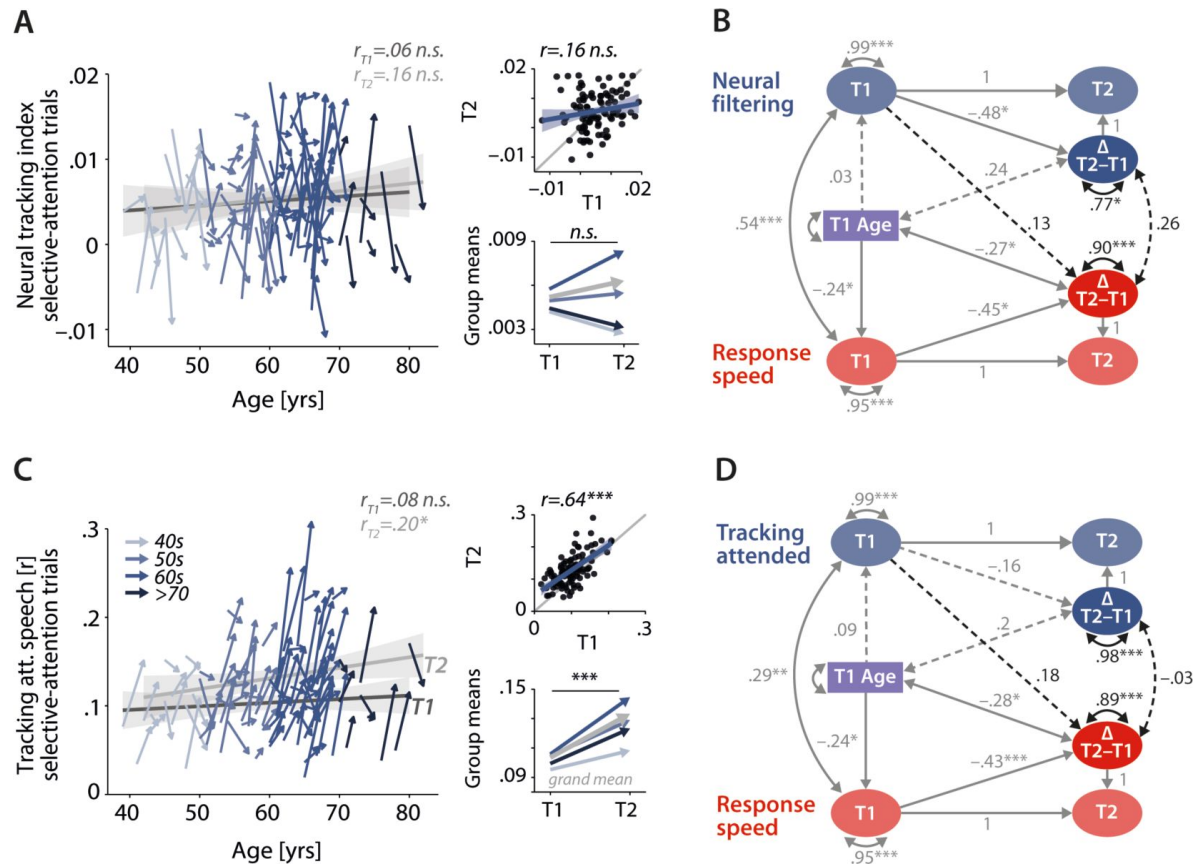


Figure 6.

Control analyses corroborate the independence of neural and behavioural trajectories under selective attention.

Cross-sectional and longitudinal change in neural filtering (A) and neural tracking of attended speech (C) averaged across selective-attention trials, only. Coloured vectors (colour-coding four age groups for illustrative purposes, only) in the left subpanels show individual T1-T2 change along with the cross-sectional trend plus 95% confidence interval (CI) separately for T1 (dark grey) and T2 (light grey). Top right, correlation of T1 and T2 as measure of test-retest reliability along with the 45° line (grey) and individual data points (black circles). Bottom right, mean longitudinal change per age group and grand mean change (grey). B, D Latent change score model (LCSM) relating two-year changes in neural filtering (B) /neural tracking (D) strength to changes in response speed. Black arrows show the paths or covariances of interest that were freely estimates, grey arrows show paths that were freely estimated or fixed as part of the structural model but did not relate to the main research questions. Solid arrows indicate statistically significant effects, dashed arrows reflect non-significant paths. All estimates are standardised. *** $p < .001$, ** $p < .01$, * $p < .05$.

predictive at all of how their attentive listening ability will develop in the future. We here have addressed these questions leveraging a new representative prospective cohort sample of healthy middle-aged to older listeners.

Over two years from T1 to T2, individuals' hearing ability worsened as expected^{23–25}. Their listening performance, however, stayed stable. In addition, an individual's baseline (T1) neural filtering strength proved to be a strikingly poor indicator of their future (T2) level of neural filtering. On the other hand, bolstering previous results, neural filtering reliably supported listening behaviour within the same session—both at T1 and T2—at two levels of granularity: Individuals with generally stronger neural tracking of target vs. distractor speech performed the listening task on average more accurately and faster. They were also more likely to respond correctly in a given trial with relatively stronger neural filtering in that trial.

Crucially, however, momentary states of neural functioning were not predictive of future behavioural change, and the dynamics of longitudinal change at the neural and behavioural level appear to follow largely independent trajectories. Notably, these key findings hold for different definitions of neural-attentional filtering, and under different spatial-attention conditions (see “Control analyses” in Results).

Neural filtering fidelity as a trait-like neural marker of individual attentive listening ability?

In recent years, the enhanced representation of behaviourally relevant sounds via their prioritised neural tracking has been reported in numerous listening studies investigating different acoustic environments, participant populations, and stages of auditory processing^{12,13,34–39}.

This neural signature is commonly interpreted as a neural instantiation of selective auditory attention in the service of successful speech comprehension. Note however that its link to behaviour is not always explicitly established^{12–14,40,41}. Given how robustly the current data show the enhanced neural tracking of attended vs. ignored speech at the group level, the weak reliability of individual neural filtering strength may come as a surprise. At the same time, stronger neural filtering was reliably linked to better behavioural performance within both T1 and T2.

How can these two findings be reconciled? Based on the current and previous results, what may be concluded about the role of neural filtering as a potential neural marker of individual attentive listening ability? Previous studies on attention-guided neural speech tracking have not provided any direct evidence on the temporal stability of neural filtering nor on its relationship with behaviour. Studies on related neural signatures such as speech-aligned auditory brainstem responses, or the entrainment of auditory cortical activity to rhythmic (non- speech) stimulation reported moderate to high reliability^{42–45}. However, these studies have (i) investigated the temporal stability across sessions spaced only days or weeks apart, (ii) focused on younger normal-hearing populations, or (iii) quantified the neural encoding of speech or non-speech stimulation that involved less or no attentional control. In contrast, we here explicitly focused on a definition of neural filtering that incorporated the neural tracking of attended and ignored speech to highlight the dependence of communication success on the attention-guided differential neural encoding of relevant vs. irrelevant input. These differences render a direct comparison to our approach difficult, but there is reason to consider a model-derived, latent representation of neural filtering as employed here the more generalisable metric⁴⁶. Leaving the latent-variable framework aside, it is worth emphasizing that the neural tracking of attended speech alone proved to be substantially more stable over time, with a re-test reliability in the $rtt = .6$ range.

Our current results broadly align with a view of attention-guided neural speech tracking as a form of ‘neural entrainment in the broad sense’ that reflects a listener’s neural attentional state^{10,47,48}.

Under this interpretation, the stable link of neural filtering to listening behaviour at different levels of granularity is noteworthy: An individual’s ability to exert top-down selective attention to prioritize the neural encoding of behaviourally relevant information is far from stable but it fluctuates at different time scales^{49–52}. This entails that at a longer time scale, here captured by two distinct measurement timepoints, aging individuals will differ with respect to their overall level of neural filtering and associated listening behaviour. Their level of neural and behavioural functioning will differ from other individuals’ levels at the same timepoint but also from their own level at a different timepoint. Moreover, a listener’s behavioural outcome at either timepoint is not only shaped by their broad neural attentional state with which they enter a communication situation. It is also critically influenced by short-term fluctuations in neural filtering strength around their current overall level of neural functioning (see **Fig. 4C**)¹². As our control analyses have revealed, these fluctuations are more pronounced if neural filtering is defined by the tracking of both attended and ignored speech rather than by attended-speech tracking alone.

What does this mean for the potential translational value of neural tracking? Their highly dynamic nature gives neural tracking-based metrics value as online neural indicators of a listener’s momentary attentional focus. As such, they could serve as critical neural read-outs in novel brain-computer interfaces such as neurally-steered hearing aids^{53–55}. At the same time, however, their non-stationarity over time limits their potential as translational targets for diagnosis and therapeutic intervention^{45,56,57}.

Individual trajectories of listening behaviour cannot be explained by changes within a single domain

As a second central query of the current study, we went beyond the establishment of robust brain-behaviour relationships and directly probed the potency of neural filtering to predict behavioural change over time¹⁵. We asked whether individual trajectories of listening behaviour could be predicted by past levels of neural filtering, or by co-occurring changes in neural filtering.

Past studies have observed enhanced cortical speech tracking in aging compared to young adults. This suggests a compensatory role of increased speech- brain coupling to counteract the deleterious effect of age or of hearing loss on speech comprehension^{41,58–62}. As a corollary of this relationship, typically observed cross- sectionally, one might expect an individual’s neural filtering strength to be connected not only to present but also to future trajectories of listening behaviour.

Such relationships, if shown to be causal, would provide the strongest evidence for the role of neural speech tracking as a neural compensatory mechanism supporting communication success³⁰. However, when analysed in our longitudinal sample of aging listeners, irrespective of the precise definition of neural filtering, we did not find evidence for a predictive role of neural filtering despite supporting brain-behaviour links observed within each time point. What are potential reasons for this absent connection (**Fig. 5**)?

One obvious explanation, both in statistical and substantive terms, may lie in the low retest reliability of our neural filtering metric as discussed above. Analytically, however, we were able to mitigate this problem by adopting a modelling approach which effectively removes the influence of measurement error²² (shown in **Fig. 5A**). Still, individual change in response speed could only be predicted by an individual’s baseline speed and age but not by baseline neural filtering nor by its longitudinal change. Moreover, our control analysis of attended-speech tracking

provided additional empirical support for the absence of such a predictive link despite the neural metric's moderate manifest-level reliability. These findings call for a more substantive explanation that goes beyond methodological details.

While most desirable from a translational perspective and a core quest in the cognitive neuroscience of aging, predicting change in cognitive functioning, here listening behaviour, from baseline or longitudinal change in brain function or structure is a non-trivial endeavour. Connecting individual trajectories of neural or cognitive functioning goes beyond the establishment of domain-specific age trends^{63,64}. It also goes beyond the mere extrapolation of (age-independent) brain-behaviour relationships observed at a given moment^{65,66}. Indeed, empirical evidence—and to some degree also theoretical grounds—for robust brain-behaviour baseline-change or change-change associations are limited¹⁸.

Most empirical studies reporting such significant cross-domain change-change correlations have in fact connected behavioural change to alterations in brain structure rather than brain function^{67–72}. Focusing on structural change may be advantageous: not only can structural feature be quantified more directly and reliably, they also follow systematic age-dependent trajectories thereby providing clearer causal pathways for ensuing behavioural change^{73,74}. Still, less than half of the studies testing such cross-domain associations have indeed observed them¹⁸. From the perspective of theoretical models of neurobiological and cognitive aging^{75–77}, the absence of correlated trajectories of neural and cognitive functioning may indeed be the more expected result. These models highlight the multifaceted nature of healthy cognitive aging in which environmental variables, neurobiology, and cognition are dynamically interrelated⁷⁸. Neural compensatory mechanisms, such as the neural filtering correlate targeted here, are thought to offset structural decline but are themselves influenced by a number of factors. This leads to increased inter-individual variability that may circumvent the emergence of group-level correlated change relationships.

Additionally, it is important to bear in mind that the behavioural outcome of interest, that is, the speed and accuracy with which an aging individual solves a difficult listening situation, involves the orchestration of different perceptual and cognitive processes^{79,80}. We here focused on one candidate neurobiological implementation of an auditory attentional filter to help explain inter-individual differences in listening behaviour and its lifespan trajectories. Yet, aging individuals may rely on different alternative neural or cognitive strategies⁸¹. A complete understanding of inter-individual differences in listening behaviour in aging adults will therefore depend on a number of different factors among which the attention-modulated tracking of relevant speech constitutes one, potentially necessary, but not sufficient neural correlate^{7,10,12,82,83}.

Not least, there are a number of methodological choices that might constrain the conclusions afforded by of our current study. First, the current study was limited to two distinct timepoints spaced only two years apart. This limits the ability to model linear as well as non-linear dynamics of change. Second, it also does not allow the separation of distinct patterns of change co-occurring at the same time: one continuous, constant change with age along with a separate process in which relative change is proportional to the level observed at prior timepoints⁸⁴. Third, we here focus on a single dichotic listening task from which both neural and behavioural functioning is derived. We therefore cannot assess whether the observed pattern would generalize to other listening tasks. Lastly, denser sampling across a longer time interval would have also increased statistical power to detect correlated change⁸⁵. It would have also allowed to more directly test hypotheses on causal pathways by which change in the neural domain should precede change in the behavioural domain.

The conclusion stands, though, that individual trajectories in listening behaviour cannot be explained by longitudinal change along a single dimension. Instead, a better understanding of the influences shaping individual listening behaviour across the adult lifespan will critically rely on

uncovering the relative contribution and age-dependent dynamics of sensory, neural, and cognitive factors.

Conclusion

The results presented here support the role of attention-guided neural filtering as a readout of an individual's neural attentional state. At the same time, the state-like nature of neural tracking-based metrics limits their translational potential as predictors of longitudinal change in listening behaviour over middle to older adulthood. Our data caution against explaining audiology-typical listening performance solely from changes in aspects of neural functioning, as listening behaviour and neural filtering ability follow largely independent developmental trajectories. Our results critically inform translational efforts aimed at the preservation and restoring of speech comprehension abilities in aging individuals.

Materials and Methods

Data collection

The analysed data are part of a large-scale longitudinal study on the neural and cognitive mechanisms supporting adaptive listening behaviour in a prospective cohort healthy middle-aged and older adults ("The listening challenge: How ageing brains adapt (AUDADAPT)"; https://cordis.europa.eu/project/rcn/197855_en.html). This project encompasses the collection of different demographic, audiological, behavioural, and neurophysiological measures across initially two time points spaced approximately two years apart. The analyses carried out on the data aim at relating adaptive listening behaviour to changes in different neural dynamics. Given the longitudinal nature of the current study, all procedures concerning data collection, as well as EEG recording and analysis are identical to those detailed in our recently published analysis of T1 data using the same experimental paradigm (see Fig. S6)¹².

Participants and procedure

We here report on a total $N = 105$ right-handed German native speakers (median age at timepoint 2 (T2) 63 years; age range 39–82 years; 61 females) who underwent audiological, behavioural, and electrophysiological (EEG) assessment at two separate timepoints. On average, the measurement timepoints were spaced $23.2 (\pm \text{SD } 4.0)$ months apart.

At timepoint 1 (T1), we had screened a total of $N = 184$ participants. Included participants had normal or corrected-to-normal vision, and did not report any neurological, psychiatric, or other disorders. They were also screened for mild cognitive impairment using the German version of the 6-Item Cognitive Impairment Test (6CIT⁸⁶) and the MoCA⁸⁷). Only participants with normal hearing or age-adequate mild-to-moderate hearing loss were included (see Fig. 2B for individual audiograms at T1). Handedness was assessed using a translated version of the Edinburgh Handedness Inventory⁸⁸. As a result of the initial screening procedure, 17 participants were excluded prior to EEG recording due to a medical history or non-age-related hearing loss. Three participants dropped out of the study prior to EEG recording and an additional 9 participants were excluded from analyses after EEG recording due to incidental findings after structural MR acquisition ($N = 3$), or due to EEG data quality issues ($N = 9$). Again, all detailed criteria are to be found in ref.¹².

At T2, $N = 115$ participants returned for follow-up measurements. All individuals passed the repeat screening procedures identical to those at T1. Ten participants had to be excluded from the analyses reported here: Three participants had dropped out prior to EEG recording, three

participants were excluded due to EEG data quality issues, and four participants because their EEG data had been excluded at T1. This resulted in a final longitudinal sample of $N = 105$ individuals.

Dropout at T2 could not be predicted from participants' T1 age, hearing loss, behavioural performance (accuracy, speed) or neural filtering strength (all $p > .13$). This indicates that compared to the full T1 cohort reported on in previous studies^{12,19} our reduced longitudinal sample was not biased in terms of sensory, cognitive, and neural functioning.

At each measurement timepoint, participants underwent detailed pure-tone and speech audiometric measurements, along with an extensive battery of cognitive tests and personality profiling (see ref.⁸¹ for details). On a separate day, we recorded participants' electroencephalogram (EEG) during rest (5 min each of eyes-open and eyes-closed measurements) followed by six blocks of the same dichotic listening task (see **Fig. 2C** and ref¹² for details).

Participants gave written informed consent and received financial compensation (10€ per hour). Procedures were approved by the ethics committee of the University of Lübeck and were in accordance with the Declaration of Helsinki.

Dichotic listening task

At each timepoint, participants performed a previously established dichotic listening task²⁰. We provide full details on trial structure, stimulus construction, recording and presentation in our previously published study on the first ($N = 155$) wave of data collection (but see also Fig. S6)¹².

In short, in each of 240 trials, participants listened to two competing, dichotically presented five-word sentences spoken by the same female speaker. They were probed on the sentence-final noun in one of the two sentences. Participants were instructed to respond within a given 4 s time window beginning with the onset of a probe screen showing four alternatives. They were not explicitly instructed to respond as quickly as possible. The probe screen showed four alternative words presented either on the left or right side of the screen, indicating the probed ear. Two visual cues preceded auditory presentation. First, a spatial-attention cue either indicated the to-be-probed ear, thus invoking selective attention, or did not provide any information about the to-be-probed ear, thus invoking divided attention. Second, but irrelevant to the current study, a semantic cue specified a general or a specific semantic category for the final word of both sentences, thus allowing to utilize a semantic prediction. Cue levels were fully crossed in a 2×2 design and presentation of cue combinations varied on a trial-by-trial level. All participants listened to the same 240 sentence pairs at each of the two measurement timepoints. The order of sentence pair presentation was randomised for each participant and at each timepoint.

To account for differences in hearing acuity within our group of participants, all stimuli were presented 50 dB above the individual sensation level.

EEG recording and analysis

The approach for EEG recording, pre-processing and subsequent analysis is identical to the procedures carried out for T1 data collection and analysis¹².

In short, 64-channel EEG data were recorded, cleaned for artifacts using a custom ICA-based pipeline, down-sampled to 125 Hz, filtered between 1–8 Hz, and cut into single-trials epochs covering the presentation of auditory stimuli. Following source-localisation via beamforming, we focused on auditory cortical activity to train and test decoding models of attended and ignored speech using cross-validated regularised regression. Models were trained on selective-attention trials, only, but then also tested on divided-attention trials. As results, we obtained single-trial reconstruction accuracy (Pearson's r) estimates as metrics of the degree of attended and ignored neural speech tracking, respectively.

We then calculated a neural filtering index across the entire sentence presentation period. The index quantifies the difference in neural tracking of the to-be-attended and of the to-be-ignored sentence (Neural filtering index = $(r_{\text{attended}} - r_{\text{ignored}}) / (r_{\text{attended}} + r_{\text{ignored}})$), and thus indexes the strength of neural filtering at the single-trial level. Positive values indicate successful neural filtering in line with the behavioural goal.

The EEG analyses were carried out in Matlab 2016b using the Fieldtrip toolbox (v. 2017-04-28), the Human Connectome Project Workbench software (v1.5), FreeSurfer (v.6.0), and the multivariate temporal response function (mTRF) toolbox (v1.5)⁸⁹.

Behavioural and audiological data analysis

We evaluated participants' behavioural performance in the listening task with respect to accuracy and response speed. For the binary measure of accuracy, we excluded trials in which participants failed to answer within the given 4-s response window ('timeouts'). Spatial stream confusions, that is trials in which the sentence-final word of the to-be-ignored speech stream were selected, and random errors were jointly classified as incorrect answers. The analysis of response speed, defined as the inverse of reaction time, was based on correct trials only.

We defined participants' hearing acuity as their pure-tone average (PTA) composed of (air-conduction) hearing thresholds at the frequencies of .5, 1, 2, and 4 kHz. Individual PTA values were then averaged across the left and right ear.

Statistical analysis

For statistical analyses focused on between-participant ('trait-level') effects, behavioural performance metrics and neural filtering index values were averaged across all trials and experimental conditions to arrive at one trait-level estimate per participants. This approach was also motivated by previous results based on T1 data: Here, we had observed that stronger neural speech tracking led to overall faster and to more trials with accurate responses irrespective of the specific cue-cue condition¹². Accuracy was logit-transformed for statistical analysis but expressed as proportion correct for illustrative purposes. Similarly, for more intuitive interpretation, we reversed the sign of PTA values for higher values to correspond to better hearing ability. All analyses were performed in R (v.4.2.2)⁹⁰ using the packages lme4 (v.1.1-31)⁹¹, mediation (v4.5.0)⁹², lavaan (v0.6-12)⁹³, and OpenMx (v2.21.8)⁹⁴.

Linear mixed-effect modelling

As the first step of our three-part analysis approach, we applied general linear mixed-effect models to tested for cross-sectional and longitudinal changes in trait-level sensory acuity, neural filtering, and listening performance. These models included age, timepoint and their interaction as fixed effect regressors and allowed random participant-specific intercepts.

In the second step of the analysis, we also aimed at replicating the previously observed single-trial state-level relationship of neural filtering and accuracy. To this end, we applied a single generalised linear mixed-effects model (binomial distribution, logit link function) on single-trial data of both T1 and T2. This model represents an adapted version of the brain-behaviour model reported in Tune et al. (2021)¹². In short, we included all experimental manipulation predictors, as well as age, hearing acuity and neural filtering metrics. We omitted previously shown to be non-significant higher-order interactions and additionally included interactions by timepoint to directly test for any longitudinal change in the effect of neural filtering on behaviour.

To tease apart state-level (i.e., within-participant) and trait-level (i.e., between-participants) effects, we included two separate neural regressors: For the between-participant effect regressor, we averaged single-trial neural filtering values per individual across all trials. By contrast, the

within-participant effect of interest was modelled by the trial-by-trial deviation from the subject-level mean⁹⁵. We included participant- and item-specific random intercepts as well as random slopes for the effect of the spatial-attention cue and the probed ear.

We used deviation coding for categorical predictors and z-scored all continuous regressors. P-values for individual model terms in general linear mixed-effect models are based on the Satterthwaite approximation for degrees of freedom, and on z-values and asymptotic Wald tests for the generalised linear mixed-effect model of accuracy⁹⁶.

Causal mediation analysis

We performed causal mediation analysis to model the direct as well as the hearing- acuity-mediated effect of age on accuracy and response speed²¹. Critically, these models also included a direct (i.e. independent of age and hearing loss) path of neural filtering on behaviour. To formally test the stability of direct and indirect relationships across time, we used a moderated mediation analysis⁹⁷. In this analysis, the inclusion of interactions by timepoint tested whether the influence of age, sensory acuity, and neural functioning on behaviour varied across time. We z-scored all dependent and independent variables, and estimated the magnitude of direct and mediated effects along with percentile-based confidence intervals on the basis of 1,000 replications.

Bayes Factor calculation

To facilitate interpretation of non-significant effects, we calculated the Bayes factor (BF) based on the comparison of Bayesian information criterion (BIC)⁹⁸. We report log Bayes Factors, with a log BF01 of 0 representing equal evidence for and against the null hypothesis; log BF01 with a positive sign indicating relatively more evidence for the null hypothesis than the alternative hypothesis, and vice versa. Magnitudes > 1 are taken as moderate, > 2.3 as strong evidence for either of the alternative or null hypotheses, respectively⁹⁹.

Latent change score modelling

In the third and final step of our analysis approach, we used structural equation modelling (SEM) to investigate the role of neural filtering as a predictor of behavioural change. We used latent change score modelling (LCSM)^{16,17} to test (i) whether an individual's T1 neural filtering strength can be predictive two-year changes in behaviour, and (ii) whether two-year changes in neural filtering and listening behaviour are systematically related. All models were specified and fitted with the R-based package lavaan⁹³ using maximum likelihood estimation.

To bring all manifest variables onto the same scale while preserving mean differences over time, we first stacked them across timepoint and then rescaled them using the proportion of maximum scale ('POMS') method^{100,101}. We assessed model fit using established indices including the χ^2 test, the root mean square error of approximation (RMSEA), and the comparative fit index (CFI)^{29,102}. Likelihood-ratio tests helped us decide whether (i) constraining individual parameters to be equal significantly decreased model fit, and (ii) individual parameter estimates were statistically significant. We report standardized parameter estimates.

In a first step, we specified separate unstructured measurement models of accuracy, speed, and neural filtering to establish factorial invariance across time using a series of tests^{17,103}. Latent variables representing each metric at T1 and T2 were constructed from the observed individual means averaged across the first and second half of the experiment, respectively. Covariances between T1 and T2 latent variables were freely estimated, residual covariances were set to be equal for the first and second half of the experiment. The factor loading and mean of the first manifest variable were set to 1 and 0, respectively, to ensure model identification¹⁰¹. Given our choice of POMS- transformation of raw values to preserve mean differences over time, the mean

of the second manifest variable had to be freely estimated to avoid model misfit. We sequentially tested for metric (i.e., identical factor loadings), strong (i.e., identical means), and strict (i.e., identical residuals) time invariance using likelihood ratio tests.

Next, for those metrics surviving factorial invariance testing, we constructed separate (i.e., univariate) change score models to test for group-level mean change following the tutorial by Kievit et al. (2018). We included a regression path from the T1 latent variable to the latent change variable to test how baseline functioning impacted the degree of longitudinal change. A likelihood ratio test of nested models (with mean change being freely estimated vs. constrained to be 0) determined the significance of group-level mean change.

In the last and final step, we constructed the bivariate LCSM that connected changes in response speed and neural filtering, which both significantly increased over time (see Fig. S2 for full model details). We modelled the covariance between the baseline neural filtering and speed, as well as the covariance between neural and behavioural change. In addition, we included a regression path from T1 neural filtering to the T1-T2 change in response speed to directly test the predictive potency of baseline neural functioning to explain behavioural change.

Following the logic of a generalised variance test¹⁰⁴[↗](#), we tested for significant inter-individual differences in neural and behavioural change. Note that such reliable variance in change is a necessary prerequisite for testing change-change correlations. In short, per domain (neural, behaviour), a likelihood ratio test compared the full model with a restricted model in which all paths pointing to or connected with the latent change variable are fixed to 0.

Finally, we asked whether neural and behavioural change covaried significantly. We ran a likelihood ratio test on the full model compared to a restricted model in which the covariance parameter was fixed to 0. For visualisation of results, we re-fit the final model using OpenMx⁹⁴[↗](#) to predict latent factor scores based on maximum likelihood.

Data and code availability

The complete dataset associated with this work including raw data, EEG data analysis results, as well as corresponding code will be publicly available under <https://osf.io/28r57/>[↗](#).

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Editors

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Reviewer #1 (Public Review):

Summary:

This study investigated behavioural performance on a competing speech task and neural attentional filtering over the course of two years in a group of middle-aged to older adults. Neural attentional filtering was quantified using EEG by comparing neural envelope tracking to an attended vs. an unattended sentence. This dataset was used to examine the stability of the link between behavior and neural filtering over time. They found that neural filtering and behavior were correlated during each measurement, but EEG measures at the first timepoint did not predict behavioural performance two years later. Further, while behavioural measures showed relatively high test-retest reliability, the neural filtering reliability was weak with an r value of 0.21. The authors conclude that neural tracking-based metrics have limited ability to predict longitudinal changes in listening behavior.

Strengths:

This study is novel in its tracking of behavioural performance and neural envelope tracking over time, and it includes an impressively large dataset of 105 participants. The manuscript is clearly written.

Weaknesses:

The weaknesses are minor, primarily concerning how the reviewers interpret their data. Specifically, the envelope tracking measure is often quite low, close to the noise floor, and this may affect test-retest reliability. Furthermore, the trajectories may be affected by accelerated age-related declines that are more apparent in neural tracking than in behaviour.

Comments on revised version:

The authors have satisfactorily addressed my previous comments and they present a strong case for the interpretation of their findings.

<https://doi.org/10.7554/eLife.92079.2.sa1>

Reviewer #2 (Public Review):**Summary:**

This study examined the longitudinal brain-behaviour link between attentional neural filtering and listening behaviour among a sample of aging individuals. The results based on the latent change score modeling showed that neither attentional neural filtering at T1 nor its T1-T2 change predicted individual two-year listening performance change. The findings suggest that neural filtering and listening behaviour may follow independent developmental trajectories. This study focuses on an interesting topic and has the potential to contribute a better understanding of the neurobiological mechanisms of successful communication across the lifespan.

Strengths:

Although research suggests that speech comprehension is neurally supported by an attention-guided filter mechanism, the evidence on their causal association is limited. This study addresses this gap by testing longitudinal stability of neural filtering as a neural mechanism upholding listening performance, potentially shedding lights on translational efforts aiming at the preservation of speech comprehension abilities among aging individuals.

The latent change score modeling approach is appropriately used as a tool to examine key developmental questions and distinguish the complex processes underlying lifespan development in brain and behaviour with longitudinal data.

Weaknesses:

Although the paper does have strengths in principle, the weaknesses of the paper are that the findings are merely based on a single listening task. Since both neural and behavioral indicators are derived from the same task, the results may be applicable only to this specific task, and it is difficult to extrapolate them to cognitive and listening abilities measured by the other tasks. Therefore, more listening tasks are required to comprehensively measure speech comprehension and neural markers.

The age span of the sample is relatively large. Although no longitudinal change from T1 to T2 was found at the group-level, from the cross-sectional and longitudinal change results (see Figure 3), individuals of different age groups showed different development pattern. Particularly, individuals over the age of 70 show a clear downward trend in both neural filtering index and accuracy. Therefore, different results may be found based on different age

groups, especially older groups. However, due to sample limitations, this study was unable to examine whether age has a moderating effect on this brain-behaviour link.

In the Dichotic listening task, valid and invalid cues were manipulated. According to the task description, the former could invoke selective attention, whereas the latter could invoke divided attention. It is possible that under the two conditions, the neural filtering index may reflect different underlying cognitive processes, and thus may differ in its predictive effect on behavioral performance. The author could perform a more in-depth data analysis on indicators under different conditions.

<https://doi.org/10.7554/eLife.92079.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

This study investigated behavioural performance on a competing speech task and neural attentional filtering over the course of two years in a group of middle-aged to older adults. Neural attentional filtering was quantified using EEG by comparing neural envelope tracking to an attended vs. an unattended sentence. This dataset was used to examine the stability of the link between behavior and neural filtering over time. They found that neural filtering and behavior were correlated during each measurement, but EEG measures at the first time point did not predict behavioural performance two years later. Further, while behavioural measures showed relatively high test-retest reliability, the neural filtering reliability was weak with an r -value of 0.21. The authors conclude that neural tracking-based metrics have limited ability to predict longitudinal changes in listening behavior.

Strengths:

This study is novel in its tracking of behavioural performance and neural envelope tracking over time, and it includes an impressively large dataset of 105 participants. The manuscript is clearly written.

Weaknesses:

The weaknesses are minor, primarily concerning how the reviewers interpret their data. Specifically, the envelope tracking measure is often quite low, close to the noise floor, and this may affect test-retest reliability. Furthermore, the trajectories may be affected by accelerated age-related declines that are more apparent in neural tracking than in behaviour.

We thank the reviewer for their supportive assessment of our work. We describe in detail how we have addressed the two main concerns raised here—neural filtering's low test-retest reliability and differences in age-related behavioural vs. neural change—in our response to the more detailed recommendations below.

To briefly summarise here:

(1) In Figure 5, we now illustrate more transparently how the employed structural equation framework helps to overcome the issue of low test-retest reliability of neural filtering as originally reported.

(2) We include two additional control analyses, one of which relates neural tracking of attended speech (featuring a moderately high T1–T2 correlation of $r = .64$ even outside of latent modelling) to behavioural change. Importantly, this analysis provides critical empirical support for the apparent independence of neural and behavioural trajectories.

(3) We more clearly describe how the latent-variable modelling strategy accounts for differences in age-related change along the neural and behavioural domain. Moreover, the results of the of 18 additional control analysis also suggest that the absence of a change-change relationship is not primarily due to differential effects of age on brain and behaviour.

Reviewer #1 (Recommendations For The Authors):

1. Figure 3:

Does the 70-year range reach a tipping point?

Is that why neural filtering drops dramatically in this age group, whereas the other groups do not change or increase slightly?

This can also be seen with behavioral accuracy to a lesser extent. Perhaps test-retest reliability is affected by accelerated age-related declines in older listeners, as was found for envelope tracking measures in Decruy et al. 2019.

We agree with the reviewer that at first glance the data seem to suggest a critical tipping point in the age range above 70 years. It is important to emphasize, however, that the four age bins were not based on equal number of data points. In fact, the >70 age group included the fewest participants, leading to a less reliable estimate of change. Together with the known observation of increasing interindividual differences with increasing age, the results do not allow for any strong conclusions regarding a potential tipping point. For the same reasons, we used the four age bins for illustrative purposes, only, and did not include them in any statistical modelling.

We did however include chronological age as a continuous predictor in latent change score modelling. Here, we modelled its influence on participants' T1 neural and behavioural status, as well as its effect on their respective change, thereby accounting for any differential (linear) effects of age on neural vs. behavioural functioning and its change.

On p.14 of the revised manuscript, we now state more clearly that the latent change score model did in fact account for the potential influence of age on the change-related relationships:

"In line with our hypotheses, we modelled the longitudinal impact of T1 neural functioning on the change in speed, and tested for a change-change correlation. Since the analyses conducted up to this point have either directly shown or have suggested that longitudinal change per domain may be affected by age, we included individuals' age as a time-invariant covariate in the final model. We modelled the influence of age on neural and behavioural functioning at T1 but also on individual change per domain. By accounting for linear effects of age on longitudinal change, we also minimize its potential impact on the estimation of change-change relationship of interest. Note that we refrained from fitting separate models per age group due to both limited and different number of data points per age group."

1. *Would good test-retest reliability be expected when the actual values of envelope tracking for attended vs. unattended speech are so low? The investigators address this by including measurement errors in the models, but I am not certain this kind adequately deals with envelope tracking values that are close to the noise floor.*

We thank the reviewer for this comment. We addressed the concerns regarding the low re-test reliability of our neural-attentional metric (and its potential impact on observing a systematic changechange relationship) in two separate ways.

The major outcome of these tests is that low re-test reliability of neural tracking is (i) not generally true, and (ii) is not the cause of the main finding, i.e., a low or absent correlations of behavioural vs. neural changes over time.

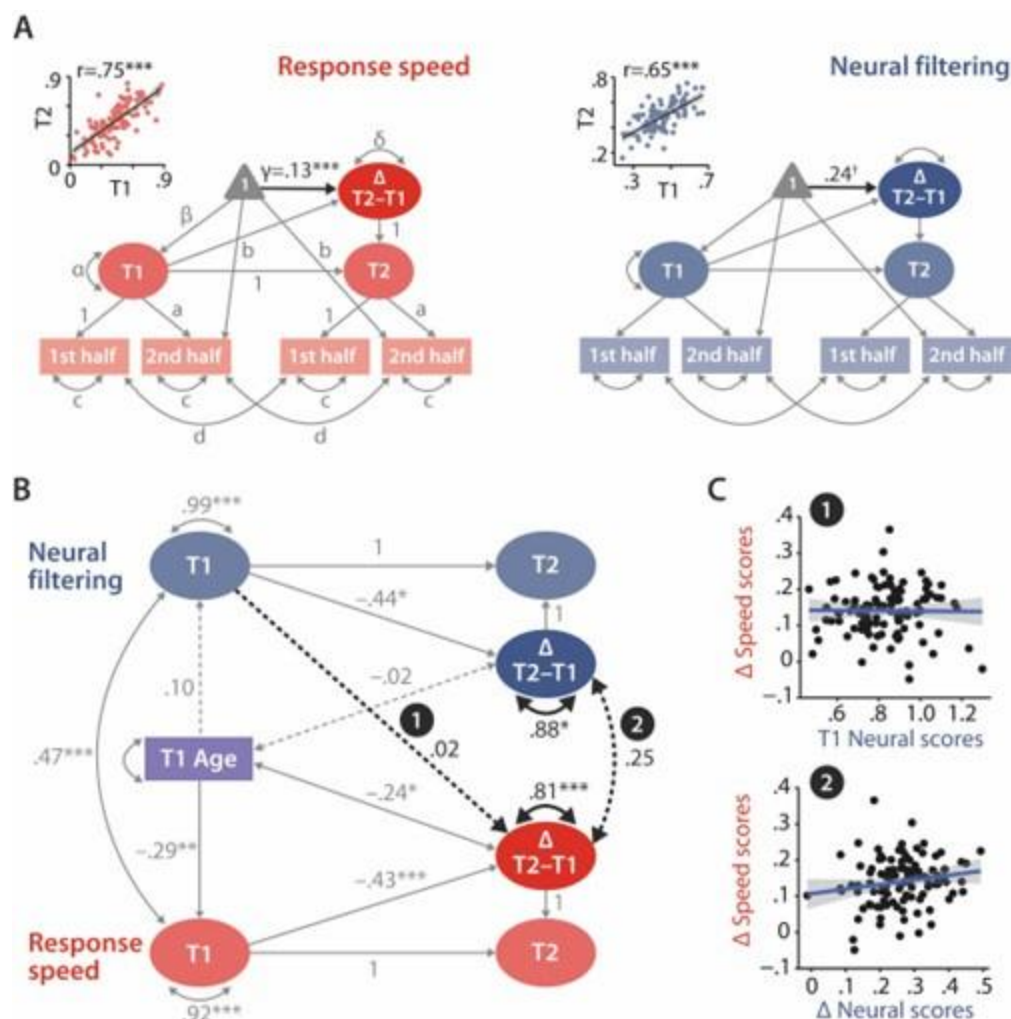
In more detail, to show how latent change score modelling improves test-retest reliability by explicitly modelling measurement error, we first extracted and correlated T1 and T2 latent factors scores from the respective univariate models of neural filtering and response speed.

Indeed, at the latent level, the correlation of T1–T2 neural filtering was moderately high at $r = .65$ (compared to $r = .21$ at the manifest level). The correlation of T1–T2 response speed was estimated as $r = .75$ (compared to $r = .71$).

Figure 5A, reproduced below for the reviewer's convenience, now includes insets quantifying these latent-level correlations over time.

Author response image 1.

Modelling of univariate and bivariate change. A Univariate latent change score models for response speed (left) and neural filtering (right). All paths denoted with Latin letters refer to freely estimated but constrained to be equal parameters of the respective measurement models. Greek letters refer to freely estimated parameters of the structural model. Highlighted in black is the estimated mean longitudinal change from T1 to T2. Scatterplots in the top left corner illustrate how capturing T1 and T2 neural and behavioural functioning as latent factors improves their respective test-retest reliability. B Latent change score model (LCSM) relating two-year changes in neural filtering strength to changes in response speed. Black arrows indicate paths or covariances of interest. Solid black arrows reflect freely estimated and statistically significant effects, dashed black arrows reflect non-significant effects. All estimates are standardised. Grey arrows show paths that were freely estimated or fixed as part of the structural model but that did not relate to the main research questions. For visual clarity, manifest indicators of the measurement model and all symbols relating to the estimated mean structure are omitted but are identical to those shown in panel A. *** $p < .001$, ** $p < .01$, * $p < .05$, $p = .08$. C Scatterplots of model-predicted factor scores that refer to the highlighted paths in panel B. Top panel shows that baseline-level neural filtering did not predict two-year change in behavioural functioning, bottom panel shows the absence of a significant change-change correlation.



Second, we ran a control analysis that includes the neural tracking of attended speech in selective attention trials rather than the neural filtering index averaged across all trials. The results are shown as part of a new main figure (and two new supplemental figures) reproduced below (see in particular Figure 6, panels C and D).

This analysis serves two purposes: On the one hand, it allows for a more direct evaluation of the actual strength of neural speech tracking as quantified by the Pearson's correlation coefficient. Note that these individual averages fall well within the to be expected range given that the neural tracking estimates are based on relatively short sentences (i.e., duration of ~2.5 sec) (O'Sullivan et al., 2014).

On the other hand, neural tracking of attended speech showed a moderately high, $r = .64$, T1–T2 correlation even outside of latent modelling. Note that the magnitude of this T1–T2 reliability is close to the short-term test-retest reliability recently reported by Panella et al. (2023). Still, when including neural tracking of attended speech in the bivariate model of change, the change-change correlation with response speed was now estimated as close to 0 ($\phi = -.03$, n.s). This observation suggests that manifest-level high re-test reliability does not necessarily improve chances of observing a significant change-change relationship.

Lastly, we would like to point out that these bivariate model results also help to shed light on the question of whether non-linear effects of age on neural / behavioural change may affect the chance of observing a systematic change-change relationship. As shown in Fig. 6C, for

neural tracking of attended speech, we observed a fairly consistent longitudinal increase across age groups. Yet, as detailed above, the change-change correlation was virtually absent.

In sum, these new results provide compelling evidence for the absence of a systematic changechange relationship.

The respective control analysis results section reads as follows, and is accompanied by Figure 6 reproduced below:

"Control analyses: The weak correlation of behavioural and neural change is robust against different quantifications of neural filtering

Taken together, our main analyses revealed that inter-individual differences in behavioural change could only be predicted by baseline age and baseline behavioural functioning, and did not correlate with contemporaneous neural changes.

However, one could ask in how far core methodological decisions taken in the current study, namely our focus on (i) the differential neural tracking of relevant vs. irrelevant speech as proxy of neural filtering, and (ii) on its trait-level characterization that averaged across different spatial-attention conditions may have impacted these results. Specifically, if the neural filtering index (compared to the neural tracking of attended speech alone) is found to be less stable generally, would this also impact the chances of observing a systematic change-change relationship? Relatedly, did the analysis of neural filtering across all trials underestimate the effects of interest?

To evaluate the impact of these consideration on our main findings, we conducted two additional control analyses: First, we repeated the main analyses using the neural filtering index (and response speed) averaged across selective-attention trials, only. Second, we repeated the main analyses using the neural tracking of attended speech, again averaged across selective-attention trials, only.

As shown in Figure 6, taken together, the control analyses provide compelling empirical support for the robustness of our main results: Linking response speed and neural filtering under selective attention strengthened their relationship at T1 ($\phi = .54$, $SE = .15$, $Dc2(df = 1) = 2.74$, $p = .1$; see. Fig 6B) but did not yield any significant effects for the influence of T1 neural filtering on behavioural change ($\beta = .13$, $SE = .21$, $Dc2(df = 1) = .43$, $p = .51$), or for the relationship of neural and behavioural change ($\phi = .26$, $SE = .14$, $Dc2(df = 1) = 3.1$, $p = .08$; please note the close correspondence to path estimates reported in Fig. 5). The second control analysis revealed a substantially higher manifest-level test-retest reliability of neural tracking of attended speech ($r = .65$, $p < .001$; Fig. 6C) compared to that of the neural tracking index. However, when linked to longitudinal changes in response speed, this analysis provided even less evidence for systematic change-related relationships: Baseline-levels of attended-speech tracking did not predict future change in response speed ($\beta = .18$, $SE = .11$, $Dc2(df = 1) = 2.73$, $p = .10$), and changes in neural and behavioural functioning occurred independently of one another ($\phi = -.03$, $SE = .12$, $Dc2(df =$

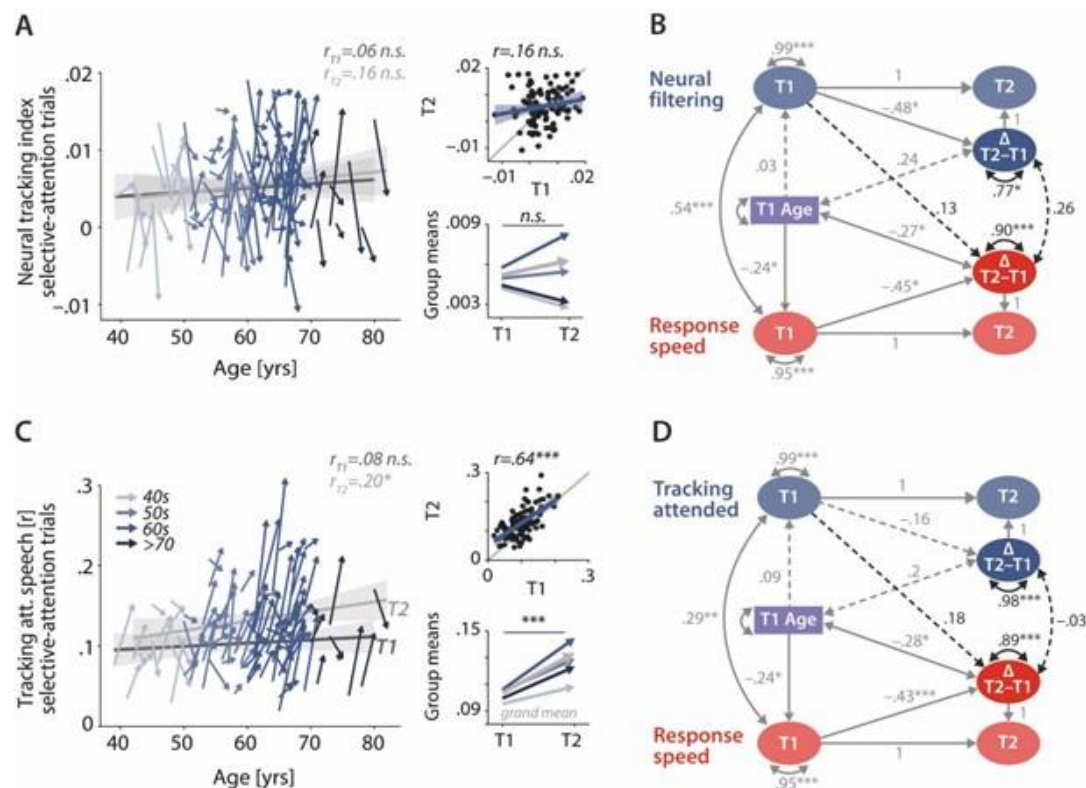
$1. = .06$, $p = .81$).

In sum, the two control analyses provide additional empirical support for the results revealed by our main analysis."

Author response image 2.

Control analyses corroborate the independence of neural and behavioural trajectories under selective attention. Cross-sectional and longitudinal change in neural filtering (A) and neural tracking of attended speech (C) averaged across selective-attention trials, only. Coloured vectors (colour-coding four age groups for illustrative purposes, only) in the left subpanels

show individual T1–T2 change along with the cross-sectional trend plus 95% confidence interval (CI) separately for T1 (dark grey) and T2 (light grey). Top right, correlation of T1 and T2 as measure of test-retest reliability along with the 45° line (grey) and individual data points (black circles). Bottom right, mean longitudinal change per age group and grand mean change (grey). B, D Latent change score model (LCSM) relating two-year changes in neural filtering (B) /neural tracking (D) strength to changes in response speed. Black arrows show the paths or covariances of interest that were freely estimates, grey arrows show paths that were freely estimated or fixed as part of the structural model but did not relate to the main research questions. Solid arrows indicate statistically significant effects, dashed arrows reflect nonsignificant paths. All estimates are standardised. *** $p < .001$, ** $p < .01$, * $p < .05$.



b

1. The authors conclude that the temporal instability of the neural filtering measure precludes its use for diagnostic/therapeutic intervention. I agree that test-retest reliability is needed for a clinical intervention. However, given the relationship with behavior at a specific point in time, would it not be a possible target for intervention to improve performance? Even if there are different trajectories, an individual may benefit from enhanced behavioral performance in the present.

We thank the reviewer for this comment. We would agree that the observation of robust between-subject (or even more desirable: within-subject) brain-behaviour relationships is a key desideratum in identifying potential interventional targets. At the same time, we would argue that the most direct way of evaluating a neural signature's translational potential is by focusing on how it predicts or is linked to individual change. In revising both the Introduction and Discussion section, we hope to now better motivate our reasoning.

Other minor comments:

1. Lines 106-107 What is the basis for the prediction regarding neural filtering?

In our previous analysis of T1 data (Tune et al., 2021), we found inter-individual differences in neural filtering itself, and also in its link to behaviour, to be independent of chronological age and hearing loss. On the basis of these results, we did not expect any systematic decrease or increase in neural filtering over time.

We rephrased the respective sentence as follows:

Since we previously observed inter-individual differences in neural filtering to be independent of age and hearing status, we did not expect any systematic longitudinal change in neural filtering.

1. Line 414: Replace "relevant" with "relevance".

Thank you, this has been corrected.

1. What was the range of presentation levels? Stimuli presented at 50 dB above individual sensation level could result in uncomfortably loud levels for people with mild to moderate hearing loss.

Unfortunately, we didn't have the means to estimate the precise dB SPL level at which our stimuli were presented. Due to the use of in-ear headphones, we did not aim to measure the exact sound pressure level of presentation but instead ensured that even if stimuli were presented at the maximally possible intensity given our hardware, this would not result in subjectively uncomfortably loud stimulus presentation levels. The described procedure estimated per individual how far the maximal sound pressure level needed to be attenuated to arrive at a comfortable and easy-to-understand presentation level.

Reviewer #2 (Public Review):

Summary:

This study examined the longitudinal brain-behaviour link between attentional neural filtering and listening behaviour among a sample of aging individuals. The results based on the latent change score modeling showed that neither attentional neural filtering at T1 nor its T1-T2 change predicted individual two-year listening performance change. The findings suggest that neural filtering and listening behaviour may follow independent developmental trajectories. This study focuses on an interesting topic and has the potential to contribute a better understanding of the neurobiological mechanisms of successful communication across the lifespan.

Strengths:

Although research suggests that speech comprehension is neurally supported by an attention-guided filter mechanism, the evidence of their causal association is limited. This study addresses this gap by testing the longitudinal stability of neural filtering as a neural mechanism upholding listening performance, potentially shedding light on translational efforts aiming at the preservation of speech comprehension abilities among aging individuals.

The latent change score modeling approach is appropriately used as a tool to examine key developmental questions and distinguish the complex processes underlying lifespan development in brain and behaviour with longitudinal data.

Weaknesses:

Although the paper does have strengths in principle, the weaknesses of the paper are that the findings are merely based on a single listening task. Since both neural and behavioral indicators are derived from the same task, the results may be applicable only to this specific task, and it is difficult to extrapolate them to cognitive and listening abilities measured by the other tasks. Therefore, more listening tasks are required to comprehensively measure speech comprehension and neural markers.

The age span of the sample is relatively large. Although no longitudinal change from T1 to T2 was found at the group-level, from the cross-sectional and longitudinal change results (see Figure 3), individuals of different age groups showed different development patterns. Particularly, individuals over the age of 70 show a clear downward trend in both neural filtering index and accuracy. Therefore, different results may be found based on different age groups, especially older groups. However, due to sample limitations, this study was unable to examine whether age has a moderating effect on this brain-behaviour link.

In the Dichotic listening task, valid and invalid cues were manipulated. According to the task description, the former could invoke selective attention, whereas the latter could invoke divided attention. It is possible that under the two conditions, the neural filtering index may reflect different underlying cognitive processes, and thus may differ in its predictive effect on behavioral performance. The author could perform a more in-depth data analysis on indicators under different conditions.

We thank the reviewer for their critical yet positive assessment of our work that also appreciates its potential to further our understanding of key determinants of successful communication in healthy aging. Please also see our more in-depth responses to the detailed recommendations that relate to the three main concern raised above.

Regarding the first concern of the reviewer about the limited generalizability of our brain-behaviour results, we would argue that there are two sides to this argument.

On the one hand, the results do not directly speak to the generalizability of the observed complex brain-behaviour relationships to other listening tasks. This may be perceived as a weakness. Unfortunately, as part of our large-scale projects, we did not collect data from another listening task suitable for such a generalization test. Using any additional cognitive tests would shift the focus away from the goal of understanding the determinants of successful communication, and rather speak more generally to the relationship of neural and cognitive change.

On the other hand, we would argue the opposite, namely that the focus on the same listening task is in fact a major strength of the present study: The key research questions were motivated by our timepoint 1 findings of a brain-behaviour link both at the within-subject (state) and at the between subject (trait) level (Tune et al., 2021). Notably, in the current study, we show that both, the state- and the trait-level results, were replicated at timepoint 2. This observed stability of results provides compelling empirical evidence for the functional relevance of neural filtering to the listening outcome and critically sets the stage for the inquiry into the complex longitudinal change relationships. We now spell this out more clearly in the Introduction and the Discussion.

Here, we briefly summarise how we have addressed the two remaining main concerns.

(1) Please refer to our response R1's comment #1 on the influence of (differential) age effects on brain and behaviour. These effects were in fact already accounted for by our modelling

strategy which included the continuously (rather than binned by age group) modelled effect of age. We now communicate this more clearly in the revised manuscript.

(2) We added two control analyses, one of which replicated the main analysis using selective attention trials, only. Critically, as shown in Figure 6, while the strength of the relationship of neural filtering and behaviour at a given timepoint increased, the key change-related relationships of interest remained not only qualitatively unchanged, but resulted in highly similar quantitative estimates.

Reviewer #2 (Recommendations For The Authors):

1. Theoretically, the relationship between brain and behavior may not be just one-way, but probably bi-directional. In this study, the authors only considered the unidirectional predictive effect of neural filtering on changes in listening task performance. However, it is possible that lower listening ability may limit information processing in older adults, which may lead to a decline in neural filtering abilities. The authors may also consider this theoretical hypothesis.

We thank the reviewer for this comment. While we did not have any specific hypotheses about influence of the behavioural state at timepoint 1 on the change in neural filtering, we ran control analysis that freely estimates the respective path (rather than implicitly assuming it to be 0). However, the results did not provide evidence for such a relationship. We report the results on p. 14 of the revised manuscript:

"We did not have any a priori hypotheses on the influence of T1 speed on the individual T1–T2 change in neural filtering. Still in a control analysis that freely estimated the respective path, we found that an individual's latent T1 level of response speed was not predictive of the ensuing latent T1–T2 change in neural filtering ($\beta = -.11$, $SE = .21$, $Dc2(df = 1) = .31$, $p = .58$)."

1. The necessity of exploring the longitudinal relationship between attentional neural filtering and listening behaviour needs to be further clarified. That is, why choose attentional filtering (instead of the others) as an indicator to predict listening performance?

We are not quite certain we understood which 'other' metrics the reviewer was referring to here exactly. But we would like to reiterate our argument from above: we believe that focusing on neural and behavioural metrics that are (i) derived from the same task, and (ii) were previously shown to be linked at both the trait- and state-level provided strong empirical ground for our inquiries into their longitudinal change-related relationships.

Please note that we agree that the neural filtering index as a measure of attention-guided neural encoding of relevant vs. irrelevant speech signals is only one potential candidate neural measure but one that was clearly motivated by previous results. Nevertheless, in the revised manuscript we now also report on the relationship of neural tracking of attended speech and listening performance (see also our response to the reviewer's comment #5 below).

Apart of this, by making the entire T1–T2 dataset openly available, we invite researchers to conduct any potential follow-up analyses focused on metrics not reported here.

1. Regarding the Dichotic listening task, further clarification is needed.

(1) The task procedure and key parameters need to be supplemented.

In a between-subjects analysis we observed, both within T1 and T2, a significant positive correlation ($r_{T1} = .33, p < .01$; $r_{T2} = .40, p < .001$) of participants' overall accuracy and response speed, speaking against a speed-accuracy trade-off. For this reason, we did not consider an integrated speed-accuracy measure as behavioural indicator for modelling.

(3) The correlation between neural filtering at T1 and T2 was weak, which may be due to the low reliability of this indicator. The generally low reliability of the difference score is a notorious measurement problem recognized in the academic community.

We fully agree with the reviewer on their assessment of notoriously noisy difference scores. It is the very reason that motivated our application of the latent change score model approach. This framework elegantly supersedes the manual calculation of differences scores, and by explicitly

modelling measurement error also removes the impact of varying degrees of reliability on the estimation of change and how it varies as a function of different influences.

While we had already detailed this rationale in the original manuscript, we now more prominently describe the advantages of the latent variable approach in the first paragraph of the Results section:

Third and final, we integrate and extend the first two analysis perspectives in a joint latent change score model (LCSM) to most directly probe the role of neural filtering ability as a predictor of future attentive listening ability. Addressing our key change-related research questions at the latent rather than the manifest level supersedes the manual calculation of notoriously noisy differences scores, and effectively removes the influence of each metric's reliability on the estimation of change-related relationships.

We also kindly refer the reviewer to our in-depth response to R1's comment #2 regarding the concern of neural filtering's low test-retest reliability and its impact on estimating change-change relationships.

1. For the latent change score model, it is recommended that the authors:
(1) Supplement the coefficients of each path in Figure 5. For details, please refer to the figures in the papers of Kievit et al. (2017, 2019)

This information has been added to Figure 5.

- (2) In Figure 5 and Figure S2, why should the two means of the observed 2nd half scores be estimated?

In longitudinal modelling, special care needs to be applied to the pre-processing/transformation of raw data for the purpose of change score modelling. While it is generally desirable to bring all variables onto the same scale (typically achieved by standardising all variables), one needs to be careful not to remove the mean differences of interest in such a data transformation step. We therefore followed the procedure recommended by Little (2013) and rescaled variables stacked across T1 and T2 using the proportion of maximum scale ('POMS') methods. This procedure, however, results in mean values per timepoint $\neq 0$, so the mean of the second half needed to be freely estimated to avoid model misfit. Note that the mean of the first half manifest variables was set to 0 (using the 'marker method'; see Little, 2013) to ensure model identification.

We have added the following more detailed description to the Method section on p. 26:

To bring all manifest variables onto the same scale while preserving mean differences over time, we first stacked them across timepoint and then rescaled them using the proportion of

maximum scale ('POMS') method99,100 (...) Given our choice of POMS-transformation of raw to preserve mean differences over time, the mean of the second manifest variable had to be freely estimated (rather than implicitly assumed to be 0) to avoid severe model misfit.

(3) The authors need to clarify whether the latent change factor in Figure 5 is $\Delta(T1-T2)$ or $\Delta(T2-T1)$?

Thank you for this comment. Our notation here was indeed confusing. The latent change factor quantifies the change from T1 to T2, so it is $\Delta(T2-T1)$. We have accordingly re-named the respective latent variables in all corresponding figures.

1. For data analysis, the author combined the trials under different conditions (valid and invalid cues) in the dichotic listening task and analyzed them together, which may mask the variations between different attention levels (selective vs. divided attention). It is recommended that the authors analyze the relationship between various indicators under different conditions.

We thank the reviewer for this comment which prompted us to (i) more clearly motivate our decision to model neural filtering across all trials, and (ii) nevertheless report the results of an additional control analyses that focused on neural filtering (or the neural tracking of attended speech) in selective-attention trials, only.

Our decision to analyse neural filtering across all spatial-attention conditions was motivated by two key considerations: First, previous T1 results (Tune et al., 2021) suggested that irrespective of the spatial-attention condition, stronger neural filtering boosted behavioural performance. Second, analysing neural filtering (and associated behaviour) across all trials provided the most direct way of probing the trait-like nature of individual neural filtering ability.

We have included the following paragraph to the Results section on p. 6 to motivate this decision more clearly:

Our main analyses focus on neural filtering and listening performance averaged across all trials and thereby also across two separate spatial-attention conditions. This choice allowed us to most directly probe the trait-like nature and relationships of neural filtering. It was additionally supported by our previous observation of a general boost in behavioural performance with stronger neural filtering, irrespective of spatial attention.

On the other hand, one could argue that the effects of interest are underestimated by jointly analysing neural and behavioural functioning derived from both selective- and divided-attention conditions. After all, it is reasonable to expect a more pronounced neural filtering response in selective-attention trials.

For this reason, we now report, in the revised version, two additional control analyses that replicate the key analyses for the neural filtering index and for the tracking of attended speech, both averaged across selective-attention trials, only: In summary, analysing neural filtering under selective attention strengthened the brain-behaviour link within a given time-point but resulted in highly similar quantitative estimated for the key relationships of interest. The analysis of attended speech tracking notably improved the neural metric's manifest-level re-test reliability ($r = .64$, $p < .001$) – but resulted in an estimated change-change correlation close to 0.

Taken together, these control analyses provide compelling support for our main conclusion that neural and behavioural functioning follow largely independent developmental trajectories.

We kindly refer the reviewer to our detailed response to R1 for the text of the added control analysis section on p. 4f. above. The additional Figure 6 is reproduced again below for the reviewer's convenience.

Author response image 4.

Control analyses corroborate the independence of neural and behavioural trajectories under selective attention. Cross-sectional and longitudinal change in neural filtering (A) and neural tracking of attended speech (C) averaged across selective-attention trials, only. Coloured vectors (colour-coding four age groups for illustrative purposes, only) in the left subpanels show individual T1–T2 change along with the cross-sectional trend plus 95% confidence interval (CI) separately for T1 (dark grey) and T2 (light grey). Top right, correlation of T1 and T2 as measure of test-retest reliability along with the 45° line (grey) and individual data points (black circles). Bottom right, mean longitudinal change per age group and grand mean change (grey). B, D Latent change score model (LCSM) relating two-year changes in neural filtering (B) /neural tracking (D) strength to changes in response speed. Black arrows show the paths or covariances of interest that were freely estimates, grey arrows show paths that were freely estimated or fixed as part of the structural model but did not relate to the main research questions. Solid arrows indicate statistically significant effects, dashed arrows reflect nonsignificant paths. All estimates are standardised. *** $p < .001$, ** $p < .01$, * $p < .05$.

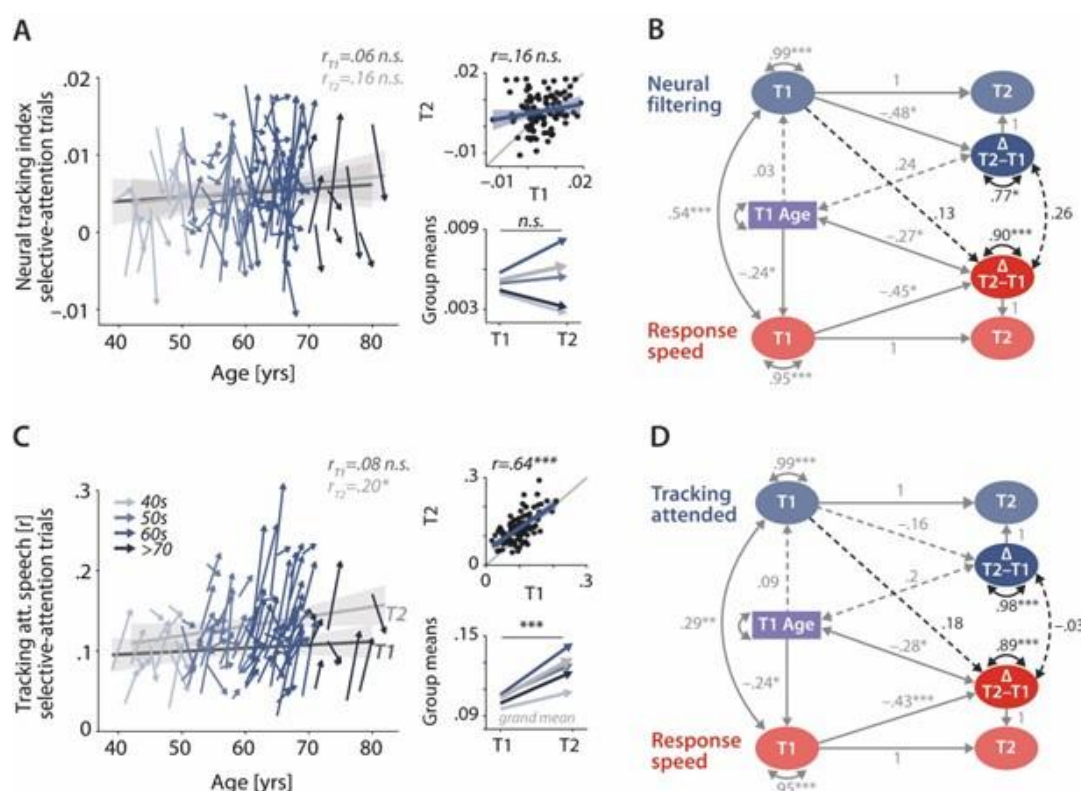
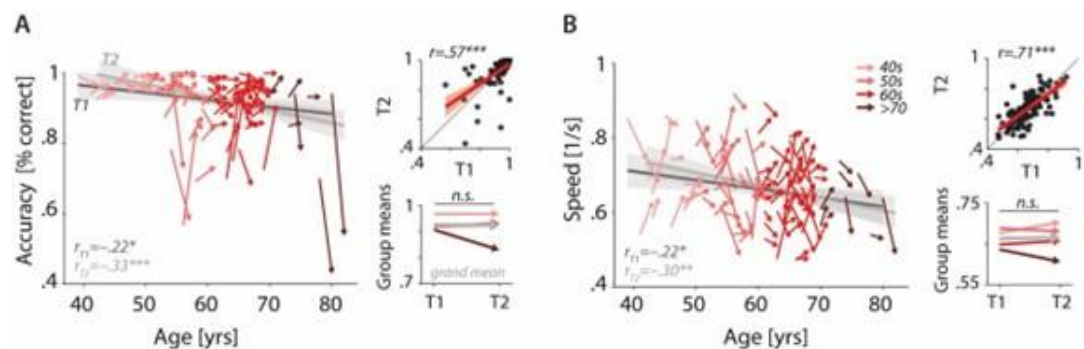


Figure 6 has also been supplemented by two additional figures showing behavioural functioning (Fig. S4) and neural tracking of ignored speech (Fig. S5) under selective-attention trials, only. These figures are reproduced below for the reviewer's convenience.

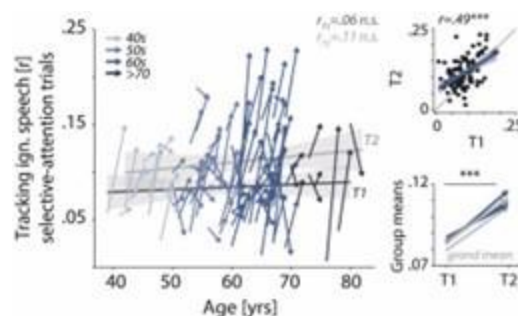
Author response image 5.

Cross-sectional and longitudinal change in listening behaviour under selective attention.



Author response image 6.

Cross-sectional and longitudinal change in neural tracking of ignored speech under selective attention.



1. As can be seen from the Methods section, there were still other cognitive tasks in this database that can be included in the data analysis to further determine the predictive validity of neural filtering.

We kindly refer the reviewer to our response to their public review and comment # 2 above where we motivate our decision to focus on manifest indicators of neural and behavioural functioning that are derived from the same task.

We believe that the analysis of several additional indicators of cognitive functioning would have distracted from our main goal of the current study focused on understanding how individual trajectories of listening performance may be explained and predicted.

1. "Magnitudes > 1 are taken as moderate, > 2.3 as strong evidence for either of the alternative or null hypotheses, respectively." Which papers are referenced by these criteria? The interpretation of BF values seems inconsistent with existing literature.

It may deserve emphasis that these are log Bayes Factors (logBF). Our interpretation of logarithmic Bayes Factors (logBF) follows Lee and Wagenmakers' (2013) classic heuristic scheme for the interpretation of (non-logarithmic, 'raw') BF10 values. We have added the respective reference to the manuscript.

Reviewer #3 (Public Review):

Summary:

The study investigates the longitudinal changes in hearing threshold, speech recognition behavior, and speech neural responses in 2 years, and how these changes correlate with each other. A slight change in the hearing threshold is observed in 2 years (1.2 dB on average) but the speech recognition performance remains stable. The main conclusion is that there is no significant correlation between longitudinal changes in neural and behavioral measures.

Strengths:

The sample size ($N > 100$) is remarkable, especially for longitudinal studies.

Weaknesses:

The participants are only tracked for 2 years and relatively weak longitudinal changes are observed, limiting how the data may shed light on the relationships between basic auditory function, speech recognition behavior, and speech neural responses.

Suggestions

First, it's not surprising that a 1.2 dB change in hearing threshold does not affect speech recognition, especially for the dichotic listening task and when speech is always presented 50 dB above the hearing threshold. For the same listener, if the speech level is adjusted for 1.2 dB or much more, the performance will not be influenced during the dichotic listening task. Therefore, it is important to mention in the abstract that "sensory acuity" is measured using the hearing threshold and the change in hearing threshold is only 1.2 dB.

We thank the reviewer for this comment. We have added the respective information to the abstract and have toned down our interpretation of the observed behavioural stability despite the expected decline in auditory acuity.

Second, the lack of correlation between age-related changes in "neuronal filtering" and behavior may not suggest that they follow independent development trajectories. The index for "neuronal filtering" does not seem to be stable and the correlation between the two tests is only $R = 0.21$. This low correlation probably indicates low test-retest reliability, instead of a dramatic change in the brain between the two tests. In other words, if the "neuronal filtering" index only very weakly correlates with itself between the two tests, it is not surprising that it does not correlate with other measures in a different test. If the "neuronal filtering" index is measured on two consecutive days and the index remains highly stable, I'm more convinced that it is a reliable measure that just changes a lot within 2 years, and the change is dissociated with the changes in behavior.

The authors attempted to solve the problem in the section entitled "Neural filtering reliably supports listening performance independent of age and hearing status", but I didn't follow the logic. As far as I could tell, the section pooled together the measurements from two tests and did not address the test-retest stability issue.

Please see our detailed response to R1's comment #2 regarding the concern of how low (manifestlevel) reliability of our neural metric may have impacted the chance of observing a significant changechange correlation.

In addition, we would like to emphasize that the goal of the second step of our analysis procedure, featuring causal mediation analysis, was not to salvage the perhaps surprisingly low reliability of neural filtering. Instead, this section addressed a different research question, namely, whether the link of neural filtering to behaviour would hold across time, irrespective of the observed stability of the measure itself. The stability of the observed

between-subjects brain-behaviour relationships was assessed by testing for an interaction with timepoint.

We have revised the respective Results section to more clearly state our scientific questions, and how our analysis procedure helped to address them:

"The temporal instability of neural filtering challenges its status as a potential trait-like neural marker of attentive listening ability. At the same time, irrespective of the degree of reliability of neural filtering itself, across individuals it may still be reliably linked to the behavioural outcome (see Fig. 1). This is being addressed next.

On the basis of the full T1–T2 dataset, we aimed to replicate our key T1 results and test whether the previously observed between-subjects brain-behaviour relationship would hold across time: We expected an individual's neural filtering ability to impact their listening outcome (accuracy and response speed) independently of age or hearing status¹². (...) To formally test the stability of direct and indirect relationships across time, we used a moderated mediation analysis. In this analysis, the inclusion of interactions by timepoint tested whether the influence of age, sensory acuity, and neural filtering on behaviour varied significantly across time."

Third, the behavioral measure that is not correlated with "neuronal filtering" is the response speed. I wonder if the participants are asked to respond as soon as possible (not mentioned in the method). If not, the response speed may strongly reflect general cognitive function or a personal style, which is not correlated with the changes in auditory functions. This can also explain why the hearing threshold affects speech recognition accuracy but not the response speed (lines 263-264).

Participants were asked to response within a given time window limited to 4 s but were not implicitly instructed to respond as quickly as possible. This is now stated more clearly in the Methods section (please also refer to our response to R2 on a similar question). It is important to emphasize—as shown in Figure 4A and Figure 5B—both at the manifest and latent variable level neural filtering (and in fact also the neural tracking of attended speech, see Fig. 6C) was reliably linked to response speed at T1 and T2. These results providing important empirical ground for the question of whether changes in neural filtering are systematically related to changes in response speed, and whether the fidelity of neural filtering at T1 represents a precursor of behavioural changes.

Moreover, an interpretation of response speed as an indicator of general cognitive function is not at all incompatible with the cognitive demands imposed by the task. As the reviewer rightly stated above, performance in a dichotic listening task does not simply hinge on how auditory acuity may limit perceptual encoding of speech inputs but also on how the goal-directed application of attention modulates the encoding of relevant vs. irrelevant inputs. We here focus on one candidate neural strategy we here termed 'neural filtering' in line with an influential metaphor of how auditory attention may be neurally implemented (Cherry, 1953; Erb & Obleser, 2020; Fernandez-Duque & Johnson, 1999).

Reviewer #3 (Recommendations For The Authors):

Other issues:

The authors should consider using terminology that the readers are more familiar with and avoid unsubstantiated claims.

For example, the Introduction mentions that "The observation of such brain-behaviour relationships critically advances our understanding of the neurobiological foundation of cognitive functioning. Their translational potential as neural markers predictive of

behaviour, however, is often only implicitly assumed but seldomly put to the test. Using auditory cognition as a model system, we here overcome this limitation by testing directly the hitherto unknown longitudinal stability of neural filtering as a neural compensatory mechanism upholding communication success."

For the first sentence, please be clear about which aspects of "our understanding of the neurobiological foundation of cognitive functioning" is critically advanced by such brain-behaviour relationships, and why such brain-behaviour relationships are so critical given that so many studies have analyzed brain-behaviour relationships. The following two sentences seem to suggest that the current study is a translational study, but the later questions do not seem to be quite translational.

The uncovering of robust between- and within-subject brain behaviour-relationships is a key scientific goal that unites basic and applied neuroscience. From a basic neuroscience standpoint, the observation of such brain-behaviour links provides important mechanistic insight into the neurobiological implementation of higher order cognition – here the application of auditory spatial attention in the service of speech comprehension. At the same time, they provide fruitful ground for translational inquiries of applied neuroscience. We therefore don't consider it contradictory at all that the current study addressed both more basic and applied/translational neuroscientific research questions.

We have rephrased the respective section as follows:

"The observation of such brain-behaviour relationships critically advances our understanding of the neurobiological foundation of cognitive functioning by showing, for example, how neural implementations of auditory selective attention support attentive listening. They also provide fruitful ground for scientific inquiries into the translational potential of neural markers. However, the potency of neural markers to predict future behavioural outcomes is often only implicitly assumed but seldomly put to the test¹⁵."

More importantly, "neuronal filtering" is a key concept in the paper but I'm not sure what it means. The authors have only mentioned that auditory cognition is a model system for "neuronal filtering", but not what "neuronal filtering" is. Even for auditory cognition, I'm not sure what "neuronal filtering" is and why the envelope response is representative of "neuronal filtering".

As spelled out in the Introduction, we define our 'neural filtering' metric of interest as neural manifestation of the attention-guided segregation of behaviourally relevant from irrelevant sounds. By terming this signature neural 'filtering', we take up on a highly influential algorithmic metaphor of how auditory attention may be implemented at the neurobiological level (Cherry, 1953; Erb & Obleser, 2020; Fernandez-Duque & Johnson, 1999).

We now provide more mechanistic detail in our description of the neural filtering signature analysed in the current study:

"Recent research has focused on the neurobiological mechanisms that promote successful speech comprehension by implementing 'neural filters' that segregate behaviourally relevant from irrelevant sounds. Such neural filter mechanisms act by selectively increasing the sensory gain for behaviourally relevant inputs or by inhibiting the processing of irrelevant inputs⁵⁻⁷. A growing body of evidence suggests that speech comprehension is neurally supported by an attention-guided filter mechanism that modulates sensory gain and arises from primary auditory and perisylvian brain regions: By synchronizing its neural activity with the temporal structure of the speech signal of interest, the brain 'tracks' and thereby better encodes behaviourally relevant auditory inputs to enable attentive listening⁸⁻¹¹."

Figure 1C should be better organized and the questions mentioned in the Introduction should be numbered.

We have revised both the respective section of the Introduction and corresponding Figure 1 in line with the reviewer's suggestions. The revised text and figure are reproduced below for the reviewer's convenience:

"First, by focusing on each domain individually, we ask how sensory, neural, and behavioural functioning evolve cross-sectionally across the middle and older adult life span (Fig. 1B). More importantly, we also ask how they change longitudinally across the studied two-year period (Fig. 1C, Q1), and whether aging individuals differ significantly in their degree of change (Q2). We expect individuals' hearing acuity and behaviour to decrease from T1 to T2. Since we previously observed inter-individual differences in neural filtering to be independent of age and hearing status, we did not expect any systematic longitudinal change in neural filtering.

Second, we test the longitudinal stability of the previously observed age- and hearing-loss-independent effect of neural filtering on both accuracy and response speed (Fig. 1A). To this end, we analyse the multivariate direct and indirect relationships of hearing acuity, neural filtering and listening behaviour within and across timepoints.

Third, leveraging the strengths of latent change score modelling^{16,17}, we fuse cross-sectional and longitudinal perspectives to probe the role of neural filtering as a precursor of behavioural change in two different ways: we ask whether an individual's T1 neural filtering strength can predict the observed behavioural longitudinal change (Q3), and whether two-year change in neural filtering can explain concurrent change in listening behaviour (Q4). Here, irrespective of the observed magnitude and direction of T1–T2 developments, two scenarios are conceivable: Intra-individual neural and behavioural change may be either be correlated—lending support to a compensatory role of neural filtering—or instead follow independent trajectories¹⁸ (see Fig. 1C)."

Author response image 7.

Schematic illustration of key assumptions and research questions. A Listening behaviour at a given timepoint is shaped by an individuals' sensory and neural functioning. Increased age decreases listening behaviour both directly, and indirectly via age-related hearing loss. Listening behaviour is supported by better neural filtering ability, independently of age and hearing acuity. B Conceptual depiction of individual two-year changes along the neural (blue) and behavioural (red) domain. Thin coloured lines show individual trajectories across the adult lifespan, thick lines and black arrows highlight two-year changes in a single individual. C Left, Schematic diagram highlighting the key research questions detailed in the introduction and how they are addressed in the current study using latent change score modelling. Right, across individuals, co-occurring changes in the neural and behavioural domain may be correlated (top) or independent of one another (bottom).

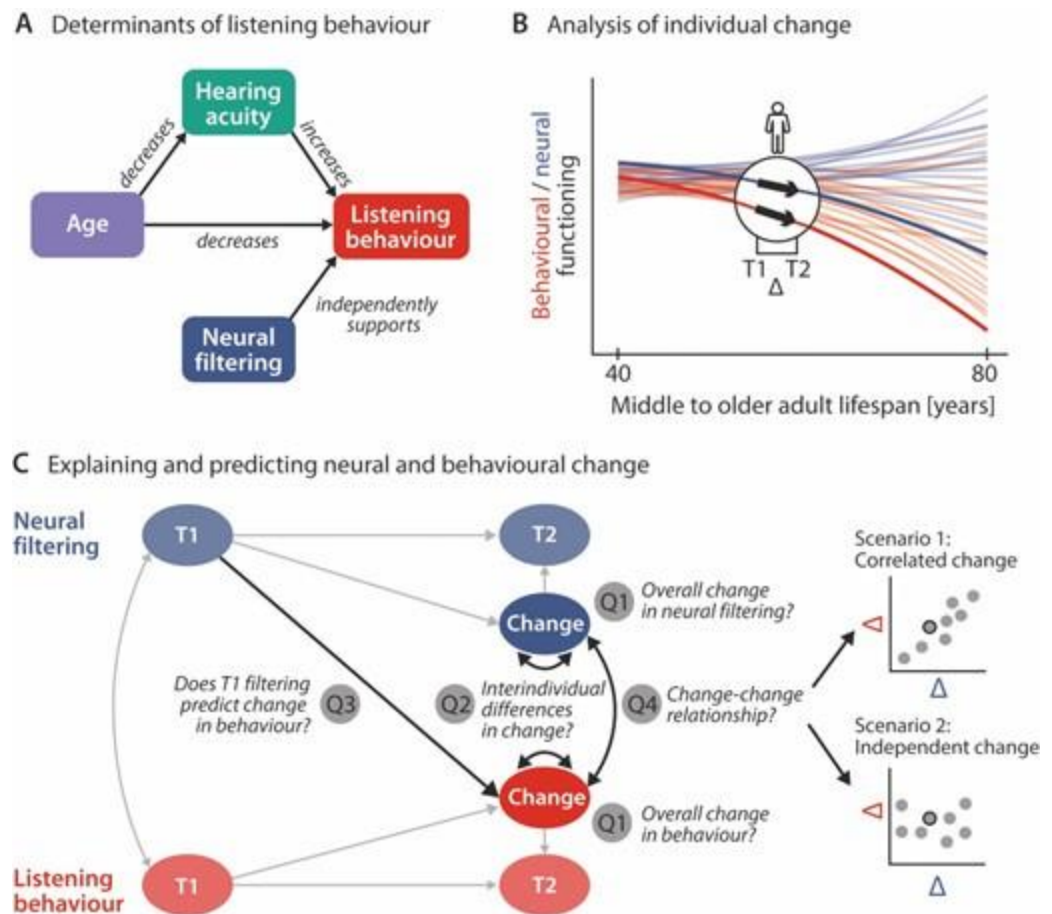
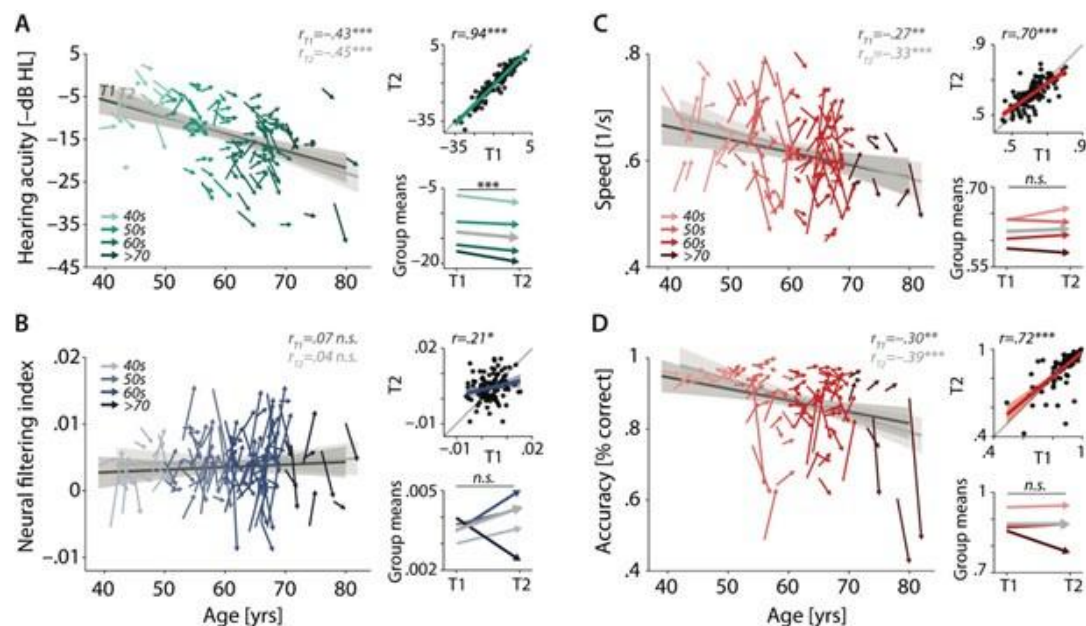


Figure 3, the *R*-value should also be labeled on the four main plots.

This information has been added to Figure 3, reproduced below.

Author response image 8.

Characterizing cross-sectional and longitudinal change along the auditory sensory (A), neural (B), and behavioural (C, D) domain. For each domain, coloured vectors (colour-coding four age groups for illustrative purposes, only) in the respective left subpanels show an individual's change from T1 to T2 along with the cross-sectional trend plus 95% confidence interval (CI) separately for T1 (dark grey) and T2 (light grey). Top right subpanels: correlation of T1 and T2 as measure of test-retest reliability along with the 45° line (grey) and individual data points (black circles). Bottom right panels: Mean longitudinal change per age group (coloured vectors) and grand mean change (grey). Note that accuracy is expressed here as proportion correct for illustrative purposes, but was analysed logit-transformed or by applying generalized linear models.



T1 and T2 should be briefly defined in the abstract or where they first appear.

We have changed the abstract accordingly.

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